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A hybrid deep feature-ensemble learning framework for robust tomato leaf disease classification using DenseNet121-EfficientNetB0 integration and SMOTE-optimize multi-classifier

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Early and accurate detection of tomato leaf diseases is essential for reducing crop losses and improving agricultural productivity. However, traditional classification approaches often struggle with subtle disease patterns, intra-class variability, and imbalanced datasets, leading to reduced diagnostic performance. This study proposes a hybrid deep feature–ensemble learning framework to improve the reliability and accuracy of automated detection of tomato leaf diseases. The proposed framework integrates two convolutional neural networks, DenseNet121 and EfficientNetB0, to extract complementary hierarchical features from tomato leaf images. To mitigate class imbalance and enhance generalisation, the Synthetic Minority Oversampling Technique (SMOTE) was applied during training. The extracted features were subsequently used to train three machine learning classifiers: Support Vector Machine (SVM), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). The predictions from these classifiers were aggregated using a soft-voting ensemble strategy to improve classification robustness and reduce model bias. Across ten independent experimental runs, the proposed framework achieved a mean accuracy of 98.72% $\pm$ 0.18, precision of 98.55% $\pm$ 0.21, recall of 98.63% $\pm$ 0.19, F1-score of 98.58% $\pm$ 0.17, and AUC of 98.21% $\pm$ 0.11. The results demonstrate that combining deep feature extraction with ensemble machine-learning classifiers significantly improves performance in tomato leaf disease classification. The framework exhibits strong generalisation across benchmark datasets, indicating its potential for deployment in intelligent crop monitoring systems and computer-aided agricultural diagnostics.

KEYWORDS
DenseNet121, EfficientNetB0, ensemble, hybrid, SMOTE

1 Introduction

Agriculture constitutes a fundamental pillar of global food security and economic stability (Pawlak and Kołodziejczak, 2020; Seppelt et al., 2022). Among cultivated crops, tomatoes rank among the most extensively grown worldwide, owing to their adaptability to a broad spectrum of moderately dry soil conditions (Cho et al., 2025). In many agricultural

regions, the majority of farmers, approximately nine out of ten, include tomatoes in their farming systems due to their high economic and nutritional value (Mango et al., 2015; Eviness et al., 2022). One of the primary challenges in tomato cultivation is the prevention and management of plant diseases, which, if not properly controlled, can result in significant yield losses or complete crop failure (Joshi et al., 2025).

The manifestation of tomato diseases can be observed through various plant components, including the leaves, stems, roots, and seeds. Among these, the leaves provide the most visible and diagnostically relevant indicators, as disease symptoms often first appear as discoloration, deformation, or spotting on leaf surfaces. However, accurate disease identification typically requires a detailed examination and analysis to determine the specific pathogen or disorder involved (Jasani et al., 2022). Incorrect diagnosis frequently leads to inappropriate treatment strategies, which may exacerbate the disease condition or fail to mitigate its spread effectively. Consequently, the timely and precise detection of tomato leaf diseases is essential for guiding suitable intervention measures and ensuring sustainable crop production. Given the substantial impact of tomato leaf diseases on crop productivity, timely and accurate diagnosis has become a crucial aspect of effective disease management in agricultural practice (Khan A. et al., 2024). Traditionally, farmers have relied on manual observation and experiential knowledge to identify disease symptoms based on visible changes in leaf coloration, texture, or morphology (Pertot et al., 2012). These conventional diagnostic practices, which often include visual inspection, expert consultation, and environmental observation, have served as the foundation for disease detection in tomato cultivation for decades. However, while these methods are accessible and cost-effective, they are inherently subjective, time-consuming, and prone to human error, particularly when distinguishing between diseases with similar visual characteristics. Consequently, the limitations of traditional diagnostic techniques underscore the need for an automated, data-driven system capable of delivering consistent, objective, and early detection of tomato leaf diseases, thereby advancing precision agriculture and fostering sustainable crop production (Kheir et al., 2025).

The ability of a deep neural network (DNN) to generalize is compromised by overfitting when it is trained on a small dataset (Sultan et al., 2025). For the precise identification of early-stage tomato leaf diseases, traditional diagnostic methods like visual inspection, experiential assessment, expert consultation, and environmental observation are frequently intrusive, time-consuming, and insufficiently sensitive (Khan M. et al., 2024; Tolba and Nabil, 2024). Furthermore, overfitting, sluggish computational performance, class imbalance, and insufficient feature extraction capacity are some of the major drawbacks of both standalone machine learning models and conventional diagnostic techniques that collectively limit their applicability in real-world agricultural settings (Haidarh et al., 2025; Kiruthika et al., 2025). The practical validity of such models is diminished, and their viability for real-time clinical applications is limited by the time required for image registration procedures, which further reduces computational efficiency.

The length of time needed for image registration procedures further lowers computational efficiency, which limits the viability of these models for real-time applications and reduces their practical validity (Zamir et al., 2025). In various plant leaf image classification tasks, deep learning methods, particularly convolutional neural networks (CNNs), have demonstrated impressive performance (Poo et al., 2023; Ray et al., 2025; Zamir et al., 2025). CNN-based models can accurately identify pathological symptoms by analysing photos of tomato leaves taken in agricultural settings (Maeda-Gutiérrez et al., 2020). Even with the significant advancements made in CNN-based tomato leaf disease classification, a number of intrinsic difficulties still exist. Previous research has mostly concentrated on individual CNN architectures, like VGG, ResNet, or Inception, without carrying out in-depth comparative studies or investigating feature-level fusion among various models (Maeda-Gutiérrez et al., 2020). Additionally, the reasoning for model selection and parameter optimization. Even though CNN-based tomato leaf disease classification has made significant strides, a number of fundamental issues still exist. Previous research has mostly concentrated on individual CNN architectures, like VGG, ResNet, or Inception, without carrying out thorough comparative analyses or investigating feature-level fusion among various models. Additionally, there is frequently a lack of clarity in the reasoning behind model selection and parameter optimization, which restricts the interpretability and transferability of published results. Few studies thoroughly evaluate model robustness across various dataset partitioning schemes or independent test sets, and problems like class imbalance are still mostly unresolved. Finding the best feature fusion and classification techniques is further hampered by the lack of thorough ablation studies and systematic assessments. These drawbacks highlight the need for a more comprehensible, dependable, and hybridized deep learning framework that incorporates complementary CNN architectures to improve diagnostic robustness and accuracy in tomato leaf disease detection. Fusing the superior feature extraction capabilities of deep networks with the effective and discriminative power of traditional machine learning or optimization algorithms, hybrid deep learning models successfully address these issues. Improved performance in the presence of imbalanced datasets, decreased computational load, and increased diagnostic precision are the outcomes of such integration (Alzahrani, 2025).

This research introduces a hybrid deep learning framework designed to achieve precise identification of tomato leaf diseases from image data. The proposed framework is designed to enhance both classification accuracy and interpretability by integrating sophisticated feature extraction techniques and utilizing transfer learning with pretrained convolutional neural network (CNN) architectures. Specifically, the framework integrates complementary feature representations derived from DenseNet121 and EfficientNetB0 to enhance its discriminative capability. To mitigate class imbalance and ensure equitable representation of all disease categories, the synthetic minority oversampling technique (SMOTE) is applied during data preparation. Furthermore, three machine learning classifiers, support vector machine (SVM), random forest (RF), and extreme gradient boosting (XGBoost), are employed as part of an ensemble strategy to boost overall classification accuracy.

The overall diagnostic robustness and reliability are then increased by implementing a soft-voting ensemble mechanism that aggregates the predictive abilities of the individual classifiers. The resulting hybrid ensemble model performs exceptionally well in correctly classifying photos of tomato leaf disease, achieving high precision and robustness that suggest its applicability for actual agricultural deployment. The model's generalization capability is further validated by evaluation on two additional datasets with different acquisition conditions.

Beyond image-level disease classification, recent research in intelligent plant diagnostics has increasingly shifted toward actionable disease assessment, including severity estimation, lesion-centric representation learning, and multimodal or multi-task diagnostic frameworks (Hemalatha and Jayachandran, 2024; Zhao et al., 2025). Approaches such as PlantPAD and PlantIF demonstrate how severity grading and symptom localization can improve interpretability and decision-making in agricultural practice (Dong et al., 2024; Wu et al., 2025). Similarly, lesion-guided and multi-task learning frameworks aim to enhance robustness under real field conditions by focusing on biologically meaningful disease regions rather than global image statistics (Fu et al., 2025; Subbarayudu and Kubendiran, 2025).

Within this evolving context, the present work focuses on robust and generalizable feature learning through dual-backbone fusion and ensemble decision-making. Rather than targeting severity estimation directly, our approach emphasizes stable disease discrimination across domain shifts, providing a reliable foundation for future extensions toward severity-aware or lesion-level diagnosis. This design choice prioritizes field deployability and generalization, aligning with the broader goal of developing practical, trustworthy plant disease diagnostic systems. All things considered, this study offers the following significant contributions.

- Creation of a hybrid deep learning framework that combines the EfficientNetB0 and DenseNet121 architectures to use multi-level feature representations for improved tomato leaf disease classification.
- The suggested hybrid model successfully combines the interpretability and generalization potential of conventional classifiers with the representational power of convolutional neural networks (CNNs) by fusing deep learning architectures with conventional machine learning algorithms.
- With the synthetic minority oversampling technique (SMOTE), class imbalance problems can be effectively mitigated, improving model generalization and lowering bias toward dominant disease classes.
- A novel soft-voting ensemble strategy is proposed to combine the predictions of multiple classifiers, enhancing the model's stability, robustness, and reliability when applied to diverse datasets.
- This study provides one of the first comprehensive benchmarking frameworks that evaluates a hybrid deep-feature ensemble model against both general-purpose deep learning architectures and mainstream agricultural AI-specific models, establishing a unified and domain-aware evaluation standard for intelligent plant disease diagnosis.

The remainder of the article is structured as follows: related work is reviewed in Section 2. The workflow diagram is explained in Section 3, Methodology. The results are explained in Section 4 presents the results and discussion, which compares CNN models for the classification of tomato plant diseases using different imaging modalities. The conclusion, which has the limitations and potential future research directions, is finally summarized in Section 5.

2 Related work

Artificial intelligence (AI) has demonstrated promising capabilities in classifying various food types using visual features (Borugadda and Kalluri, 2025). In the agricultural sector, machine learning and deep learning techniques have been widely applied for tasks such as disease detection (Benti et al., 2024; Faisal et al., 2025; Waqas et al., 2025) and yield prediction (Jabed and Murad, 2024; El –Kenawy et al., 2025; Qin et al., 2025). Specifically, deep learning models, as a subset of AI, have shown strong potential in accurately identifying tomato diseases based on their visual characteristics. Table 1 summarizes recent studies on the application of AI in tomato leaf classification.

The rapid evolution of modern technologies has significantly accelerated the expansion of artificial intelligence (AI), driving innovation across numerous sectors (Hirsch-Kreinsen, 2024; Li et al., 2024; Rashid and Kausik, 2024). Deep learning, in particular, has gained widespread application in fields such as technology, healthcare (Bajwa et al., 2021), economics (Mienye and Swart, 2024), commerce (Ahmed et al., 2025), education (Ali et al., 2024), and agriculture (Mana et al., 2025). Traditionally, disease identification depended heavily on expert observation, a method that is not only labour-intensive but also limited in its diagnostic accuracy and scalability (Islam et al., 2023; George et al., 2025). Recent research has introduced more advanced approaches to address these challenges. For example, the study in Poo et al. (2023) applied a GauGAN-based data augmentation strategy for tomato leaf disease classification using the PlantVillage dataset. Their comparison with standard augmentation techniques, such as rotation, flipping, mix-up, and cut-mix, revealed that GauGAN improved F1-scores by over 10%. Similarly, the authors in Huo et al. (2025) employed StyleGAN3 to generate high-quality synthetic images representing tomato growth stages and trained a Vision Transformer (ViT) model on the enhanced dataset. Using 2,723 images, their augmented model outperformed non-augmented baselines, including AlexNet, DenseNet50, and VGG16, in validation accuracy.

A recent study presented in Joshi et al. (2025) proposes a hybrid data augmentation strategy that expands a limited dataset of 737 tomato leaf images to 6,696 samples for disease detection using the YOLOv8n convolutional detection framework. With the enriched dataset, the model attained 96.5% mAP, 97% precision, 93.8%

TABLE 1 Summary of recent studies on the application of AI in tomato leaf classification.

Ref.	Year	Dataset used	Method	Augmentation method
Badiger and Mathew (2023)	2023	PlantVillage dataset	Deep batch-normalized eLu Alex net (DbneAlexNet)	Position and colour augmentation
Kokate et al. (2024)	2024	PlantVillage dataset	xception	Rotation augmentation
Hossain et al. (2023)	2023	PlantVillage dataset	Deep convolutional neural networks (DCNN).	–
Wagle et al. (2021)	2021	PlantVillage dataset	ResNet 101 deep learning model and transfer learning combined with softmax classification	Colour data augmentation
Nirmala and Singh (2023)	2023	PlantVillage dataset	Deep learning models (such as AlexNet, VGG16, GoogLeNet, MobileNetV2, SqueezeNet, ResNet18, ResNet50, ResNet101)	Data augmentation involves rotation in horizontal and vertical flips.
Ghodekar et al. (2023)	2023	PlantVillage dataset	NasNetMobile model	The data augmentation pipeline includes a random adjustment to contrast and brightness
Wu et al. (2020)	2020	PlantVillage dataset	GoogLeNet model.	Deep convolutional generative adversarial network (DCGAN)
Shoaib et al. (2022)	2022	PlantVillage dataset	InceptionNet model	–
Attallah (2023)	2023	PlantVillage dataset	Deep learning (DL) model that involves three compact convolutional neural networks (CNNs) with k-nearest neighbour (k-NN) and support vector machine (SVM).	–
Al-gaashani et al. (2021)	2021	PlantVillage dataset	Transfer learning and features concatenation with machine learning	
Sharmila and Natarajan (2024)	2024	Individual-owned tomato dataset	An ensemble model consists of multiple pre-trained convolutional neural networks (CNNs), incorporating the Residual network (ResNet), MobileNet, and Inception model.	Data augmentation includes rotation, flipping, zooming, and brightness adjustments.
Angulakshmi et al. (2024)	2024	PlantVillage dataset	Transfer learning-based deep learning VGG-16	Data augmentation such as rotation, flip, scaling, cropping, colour jittering, noise addition, and elastic deformation
Wang and Liu (2024)	2024	Curated dataset	A novel method called TomatoDet and Meta-ACON	–
Abouelmagd et al. (2024)	2024	PlantVillage dataset	Optimized capsule neural network (CapsNet).	Data augmentation: width data rotation, shear range, width shift range, height shift range, and zoom range.
Nawaz et al. (2022)	2022	PlantVillage dataset	Improved Faster-RCNN approach with ResNet-34.	
Gosh et al. (2025)	2025	PlantVillage dataset	Advanced deep learning technologies combining Swin Transformer, Vision Transformer, and efficientNetV2	Image augmentation techniques such as rotation, scaling, and horizontal flipping.
Gupta et al. (2024)	2024	External tomato dataset from GitHub repository	Fine-tuned efficientNetB7	–

recall, and a 95% F1-score, demonstrating the substantial impact of augmentation on improving detection accuracy. Similarly, the work evaluates multiple ResNet variants (ResNet50, ResNet18, ResNet101) across nine tomato disease classes, as described in Wagle et al. (2021). Their results compare performance on the original dataset with two augmented versions (augdata1 and augdata2). Notably, ResNet101 trained with the more extensive augmentation (augdata2) achieved significantly enhanced performance compared to training on the unaugmented images. While much of the literature emphasizes image-based disease classification, research in yield prediction also highlights the value of enriched datasets. The study in Gong et al. (2023) integrates the Tomgro biophysical model with a CNN–RNN architecture to forecast greenhouse tomato yield. Although explicit image augmentation is not a primary focus, the incorporation of additional input features and synthetic handling of environmental variability improves predictive performance. The hybrid model recorded highly accurate results, with mean ± standard deviation values of RMSE, R^2, NSE, and PBIAS of 17.69 ± 3.47 g/m^2, 0.9995 ± 0.0002, 0.9989 ± 0.0004, and 0.1791 ± 0.6837, respectively.

Badiger and Mathew (2023) addressed the problem of the accurate detection of tomato plant leaves using a deep batch-normalized eLu Alex net (DbneAlexNet) model. The experimental result obtained gives an accuracy of 92.4%, true positive rate (TPR) of 91.9%, true negative rate (TNR) of 92.2%, and false positive rate (FPR) of 0.078. Nirmala and Singh (2023) worked on the correct classification of tomato leaf diseases using the Xception model, and the results were acceptable. The issue of accurate image classification to improve results in agriculture and technology was introduced by Ghodekar et al. (2023) using deep convolutional neural networks (DCNNs), such as VGG-19, MobiNet-V2, and ResNet-50. The model was evaluated on the PlantVillage dataset, and the results produced a satisfactory performance. Also, Nirmala and Singh (2023) addressed the problem of accurate classification of tomato leaf diseases with deep learning models (such as AlexNet, VGG16, GoogLeNet,

MobileNetV2, SqueezeNet, ResNet18, ResNet50, ResNet101). The results obtained showed that all the models achieved satisfactory accuracy (exceeded > 99%) except SqueezeNet.

Ghodekar et al. (2023) developed a fast and lightweight tomato leaf disease recognition approach using the NasNetMobile architecture on the PlantVillage dataset, achieving 79.30% precision, 69.20% recall, an F1-score of 67.40%, and an accuracy of 69.20%. Attallah (2023) addressed crop health deterioration by proposing a deep learning framework that combines three compact CNN models with k-nearest neighbour (k-NN) and support vector machine (SVM), resulting in a high accuracy of 99.92%. Sharmila and Natarajan (2024) investigated tomato diseases caused by bacterial, fungal, and viral infections using a privately owned dataset and an ensemble of pretrained CNN models: ResNet, MobileNet, and Inception, achieving 98.86% accuracy. Angulakshmi et al. (2024) utilized a transfer learning strategy with the VGG-16 model to enhance early plant disease detection, reporting an accuracy of 93.7%. Nawaz et al. (2022) tackled the challenge of precise localization and identification of tomato leaf diseases using an improved Faster R-CNN with a ResNet-34 backbone, yielding strong and reliable performance. Gupta et al. (2024) further advanced tomato leaf disease classification by integrating Swin Transformer, Vision Transformer, and EfficientNetV2 models, reaching 98% accuracy. Additionally, Abouelmagd et al. (2024) proposed an optimized capsule neural network (CapsNet) to improve the detection of disease characteristics such as colour, shape, and lesion position, obtaining an accuracy of 96.39%.

Recent advances in plant disease recognition increasingly favour Transformer-based and hybrid CNN–CNN-Transformer architectures over traditional convolutional networks. Several studies demonstrate that Vision Transformers (ViT) can capture long-range spatial patterns and multi-scale cues that are critical for differentiating subtle foliar lesions. For instance, Murugavalli and Gopi (2025) show Vision Transformers (ViT) and hybrid ViT +CNN architectures achieve strong accuracy gains on leaf/disease benchmarks, especially when combined with task-specific tweaks (multi-scale attention, local spatial hierarchies, or Mixture-of-Experts). Tonmoy et al. (2025) found that mobile/lightweight ViT hybrids (e.g., Mobile-friendly Hybrid ViT, lightweight CNN +explainable Mob-Res) achieve competitive accuracy with far fewer parameters, making them attractive for on-device workflows. A parallel direction focuses on advanced augmentation strategies to mitigate overfitting to laboratory-style datasets such as PlantVillage. While standard geometric transforms remain common, recent work highlights the effectiveness of compositional augmentations such as MixUp, CutMix, AugMix, and RandAugment in improving minority-class performance and robustness under distribution shifts (Naveed et al., 2024). Newer variants such as EdgeCutMix, which mixes images along leaf-edge structures rather than random regions, have demonstrated superior lesion localization stability and improved generalization in low-sample regimes. Several recent works construct photorealistic synthetic datasets/digital twins (crop-scale simulators) and apply sim2real transfer to overcome data scarcity and field variability; these pipelines are often combined with domain-adaptation or semi-supervised learning (mean-teacher, warm-start pseudo-labelling) to bridge the synthetic field gap. Sim2real, plus a domain adaptation stage, is a promising route to get real-world generalization when field data are scarce (Mirbod et al., 2025).

In contrast to these lines of research, our method integrates dual deep feature extractors (DenseNet121 + EfficientNetB0) with a soft-voting ensemble of classical machine learning classifiers to leverage complementary texture- and shape-sensitive representations. This design aligns with prior evidence that multi-branch or multi-backbone fusion can improve stability for heterogeneous leaf datasets. However, unlike many previous studies, we incorporate (i) rigorous data leakage checks, (ii) extensive statistical testing (paired t-tests, Wilcoxon tests, effect sizes, bootstrap CIs), and (iii) deployment-focused measurements (model size, inference latency, and peak GPU memory), along with a lightweight MobileNetV2 edge variant for resource-constrained environments. Our comparison against strong modern baselines, including ViT, EfficientNetV2, MobileNet, and CNN–CNN-Transformer hybrids, positions this work within the most recent trajectory of research focused simultaneously on accuracy, robustness, and practical deployability.

3 Method

This section shows the entire process of the suggested hybrid deep feature-ensemble learning framework for tomato leaf disease classification. The pipeline includes (i) contrast-limited adaptive histogram equalization (CLAHE) for image preprocessing, (ii) DenseNet121 and EfficientNetB0 for dual-stream deep feature extraction, (iii) vector concatenation for feature fusion, (iv) SMOTE for class imbalance mitigation, and (v) an ensemble of random forest, support vector machine, and XGBoost classifiers for multi-class classification. A rigorously regulated protocol for data splitting and reproducibility is followed in every experiment.

3.1 Design rationale of the proposed hybrid framework

DenseNet121 and EfficientNetB0 were selected due to their complementary architectural properties. DenseNet121 facilitates dense feature reuse and is particularly effective for capturing fine-grained texture patterns, while EfficientNetB0 employs compound scaling to preserve global structural information with high parameter efficiency. Feature-level fusion allows the framework to leverage these complementary representations without incurring the computational cost of end-to-end retraining. Rather than relying on a single Softmax classifier, classical machine learning classifiers were employed to exploit the structured deep embeddings. The ensemble of SVM, random forest, and XGBoost balances margin-based separation, robustness to noise, and non-linear decision boundaries, resulting in improved stability and generalization. The proposed system framework is shown in Figure 1.

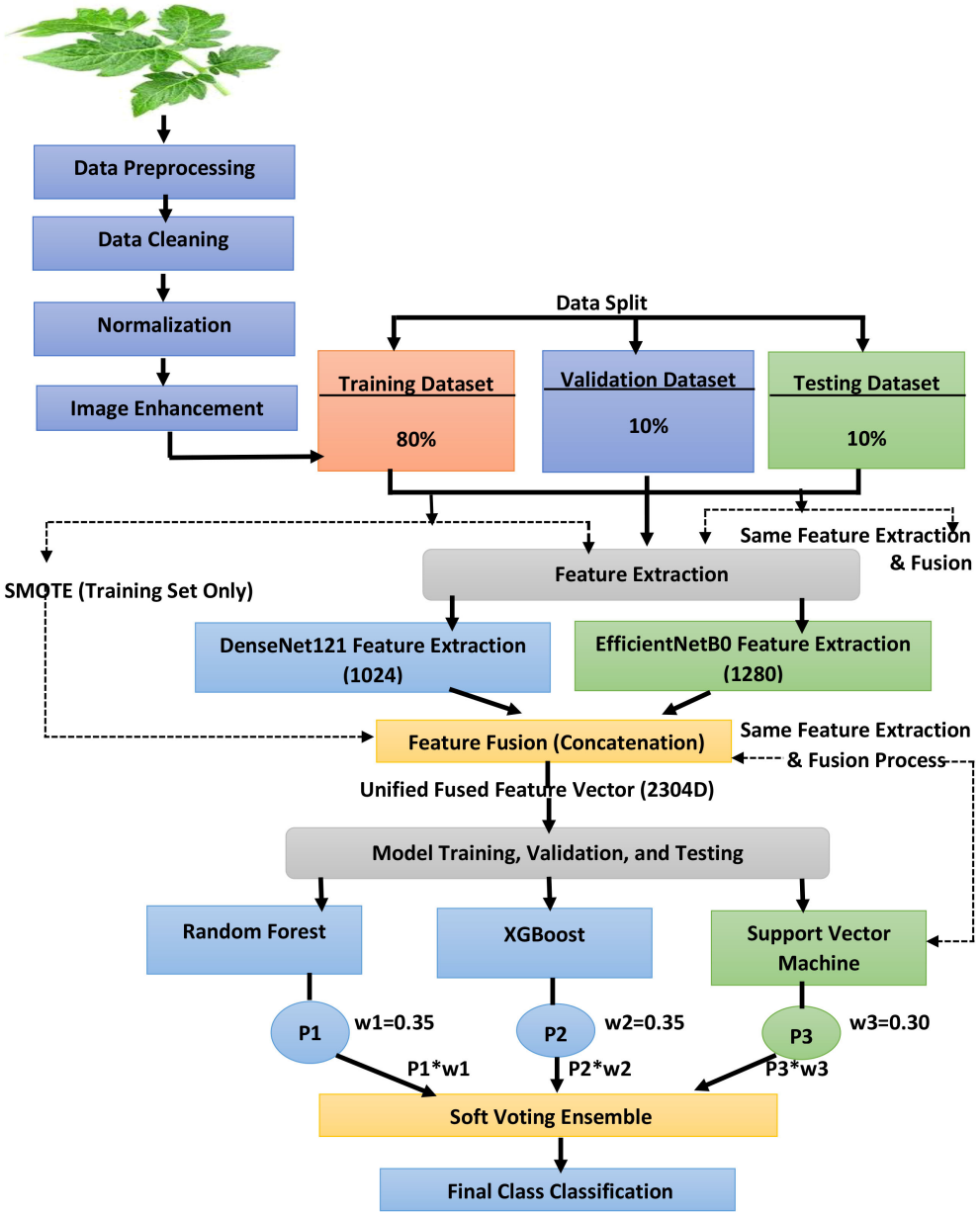


FIGURE 1
System framework for the proposed method.

3.2 Dataset and experimental split strategy

The tomato plant images used in this research were sourced from the publicly available Kaggle dataset The dataset is available at https://www.kaggle.com/datasets/emmarex/plantdisease. The dataset is distributed under an open academic license, allowing unrestricted use for research and publication purposes. To ensure a reliable assessment and avoid any possibility of data leakage, the original image files, excluding any augmented versions, were randomly partitioned into 80% for training, 10% for validation, and 10% for independent testing. All augmentation procedures and SMOTE-based oversampling were applied exclusively to the training portion of the dataset. The dataset partitions and random seed = 42 (plus an additional 10 seeds for statistical testing) were saved as index files (train_indices) to improve reproducibility. csv and val_indices. csv and test_indices (csv). Every experiment was carried out with the same software and hardware. This method guarantees a fair comparison and keeps the tests from becoming contaminated.

3.2.1 Preventing data leakage

Preventing data leakage is critical to ensuring valid model evaluation, especially in image-based plant disease classification, where augmented variants of the same image can inadvertently appear across different subsets. Leakage can artificially inflate classification performance, undermine reproducibility, and compromise scientific integrity. This study incorporates a

rigorous, multi-stage leakage prevention framework comprising controlled dataset partitioning, restricted augmentation, isolated feature resampling, and formal leakage verification experiments.

3.2.2 Controlled dataset partitioning

The dataset was first partitioned into training (80%), validation (10%), and testing (10%) sets before any augmentation or preprocessing. Split indices were stored in three immutable files: train_indices.csv, val_indices.csv, and test_indices.csv. Each entry consists of a unique image filename and the global random seed (Islam et al., 2023). All subsequent data processing stages referenced these split files to ensure reproducibility and prevent accidental reallocation. This strict partitioning ensures that images and their augmented variants are isolated within their designated subsets. A few samples of the tomato plant leaf diseases are shown in Figure 2.

3.3 Data preprocessing

The tomato plant leaf images underwent several preprocessing and augmentation steps to increase contrast, reduce noise, and facilitate robust training. The dataset was partitioned using an 80:10:10 split into training, validation, and test sets, respectively. To prevent data leakage and ensure experimental integrity, SMOTE and data augmentation were applied exclusively to the training set. The validation and test sets were processed using identical preprocessing and feature extraction pipelines without any form of resampling, augmentation, or synthetic data generation. This design ensures that model evaluation reflects true generalization performance and that no artificial data distributions influence validation or test outcomes.

To emphasize fine-grained textures, CLAHE was first applied to each grayscale tomato plant leaf image. With a tile grid size of (8,8) and a clip limit of 2.0, the CLAHE parameters were empirically chosen to achieve the best contrast enhancement without increasing noise. To balance edge preservation and smoothness, the parameters for bilateral filtering were set to diameter d = 6, σ_{Space} = 78, and σ_{Color} = 78. To prevent noise amplification, CLAHE divided the image into small, non-overlapping regions and clipped their histograms at a predetermined threshold T_i. $\hat{H}_i = \min(H_i, T)$, defines the clipped histogram $\hat{H}_i$, where H_i is the initial histogram count for bin i. As in Equation 1, all L_i histogram bins receive an equal distribution of the total access C_i.

$$\hat{H}_i^f = \hat{H}_i + \frac{C_i}{L_i} \tag{1}$$

Where, $C_i = \sum_{i=1}(L_i - 1)(H_i - T_i)^+$ and $(x)^+ = \max(x, 0)$. Subsequently, the pixel intensities are remapped by the Cumulative Distribution Function (CDF) as in Equation 2.

$$I'(x_i, y_i) = \left[\frac{CDF(I(x_i, y_i)) - CDF_{min}}{N - CDF_{min}} \times (L_i - 1)\right] \tag{2}$$

Where, N is the tile's total pixel count. This process effectively reduces noise and artefact enhancement while increasing local contrast. After that, bilateral filtering is used to minimize noise while maintaining edges. Equation 3 describes how this filter averages neighbouring pixel intensities to achieve non-linear, edge-preserving smoothing.

$$I_{filtered}(p_i) = \frac{1}{W_{p_i}} \sum_{q_i \in \Omega} G_{sa}(||p_i - q_i||) \cdot G_{ra}(|I(p_i) - I(q_i)|) \cdot I(q_i) \tag{3}$$

Where, G_{sa} and G_{ra} represent the spatial and range kernel functions, Ω denoting the neighbourhood, and W_{p_i} acting as a normalization factor. Figure 3 shows the changes in the image after applying the CLAHE and bilateral filtering.

3.3.1 Training-only augmentation pipeline

Image augmentation (horizontal flips, rotations, zooming, translations, and CLAHE transformations) was explicitly restricted to training images, as in Equation 4.

$$Augment(I), \text{ only if } I \in Training\ \ set \tag{4}$$

Validation and test images were not subjected to any augmentation or oversampling, ensuring they serve as true estimates of generalization ability. This prevents the most common source of leakage in image classification pipelines.

Transformation methods like rotation, flipping, and zooming were used to further expand the dataset and improve generalization (Wagle et al., 2021). After that, all augmented images were resized to 224 by 224 pixels, converted to RGB format, and normalized using ImageNet's mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225], as shown in Equation 5.

$$I' = \frac{I - \mu}{\sigma} \tag{5}$$

Where I stands for the pixel values. To guarantee feature compatibility with pretrained convolutional neural networks (CNNs), normalization using ImageNet statistics was used, stabilizing the learning process and enhancing convergence. To reduce the risk of overfitting, geometric data augmentation techniques, such as rotation, flipping, and zooming, were applied to introduce additional variability into the training images. DenseNet121 and EfficientNetB0 were used to extract deep features from the processed and augmented data after they were arranged into mini-batches. An ensemble of machine learning classifiers, including random forest, support vector machine (SVM), and XGBoost, were used to concatenate and classify the resulting feature representations using a soft voting mechanism.

3.4 Dual-stream deep feature extraction

The backbone of the proposed model consists of two complementary convolutional neural network architectures: DenseNet121 and EfficientNetB0. Both models were used as fixed feature extractors (transfer learning mode) by removing their fully connected classification layers and retaining the global average pooling outputs. Dual-CNN feature fusion constitutes a core methodological component of the proposed framework. Feature representations are independently extracted from DenseNet121 and EfficientNetB0, producing feature vectors of 1024 and 1280 dimensions, respectively.

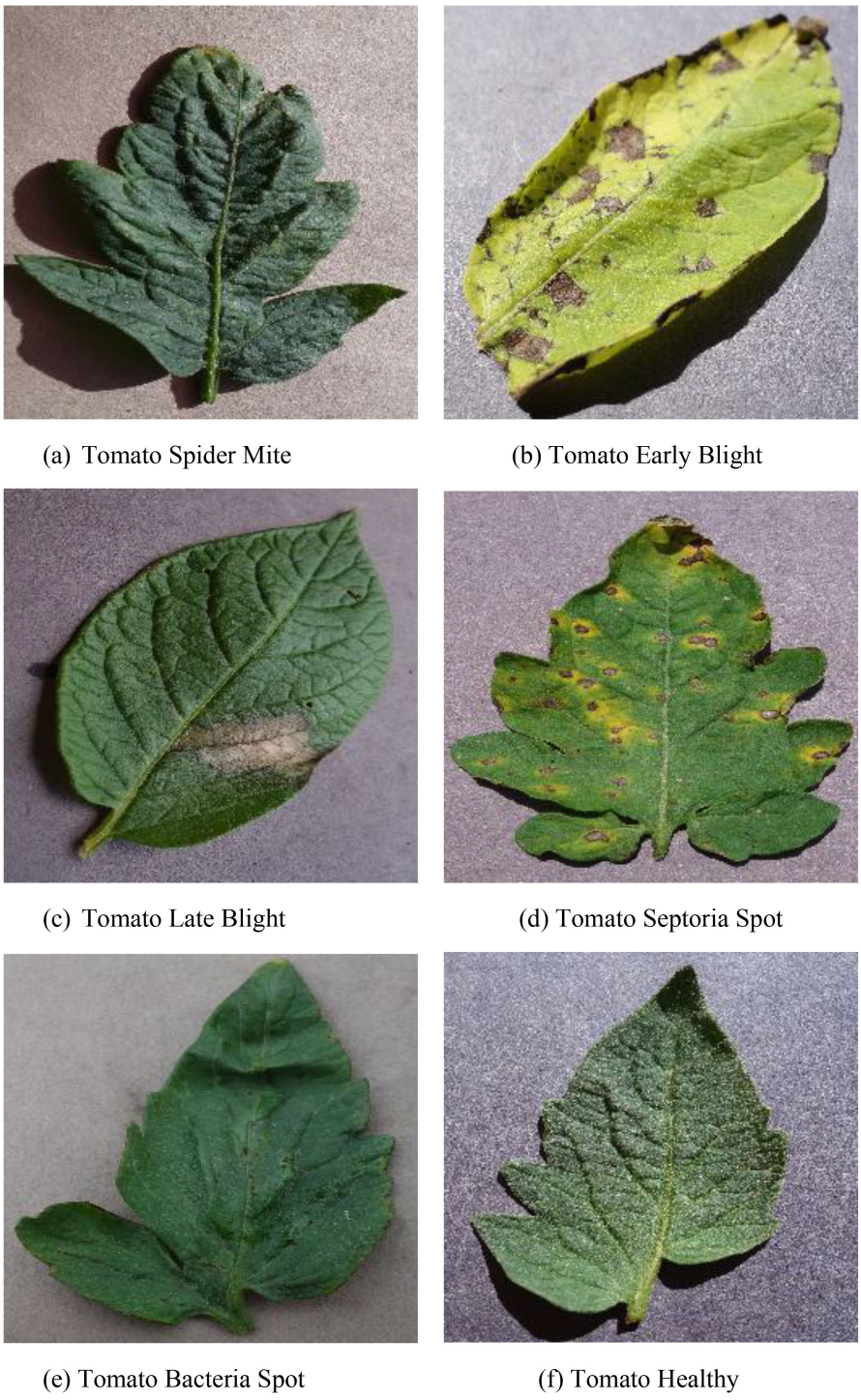


FIGURE 2
Sample of tomato leaf dataset (Haidarh et al., 2025; Kiruthika et al., 2025).

These deep representations are then combined through vector concatenation, forming a unified fused feature vector: $F_{fused} = F_{DenseNet121} || F_{EfficientNetB0}$, resulting in a 2304-dimensional fused feature space. This fused representation serves as the input basis for SMOTE resampling (training set only) and subsequent classification, enabling the model to leverage complementary hierarchical and structural features learned by both CNN architectures.

3.4.1 DenseNet121 feature extraction

DenseNet121 employs dense connectivity, where each layer receives feature maps from all previous layers. This architecture promotes feature reuse and gradient propagation, making it particularly effective for capturing texture-rich disease patterns. The output dimension of DenseNet121 features was 1024.

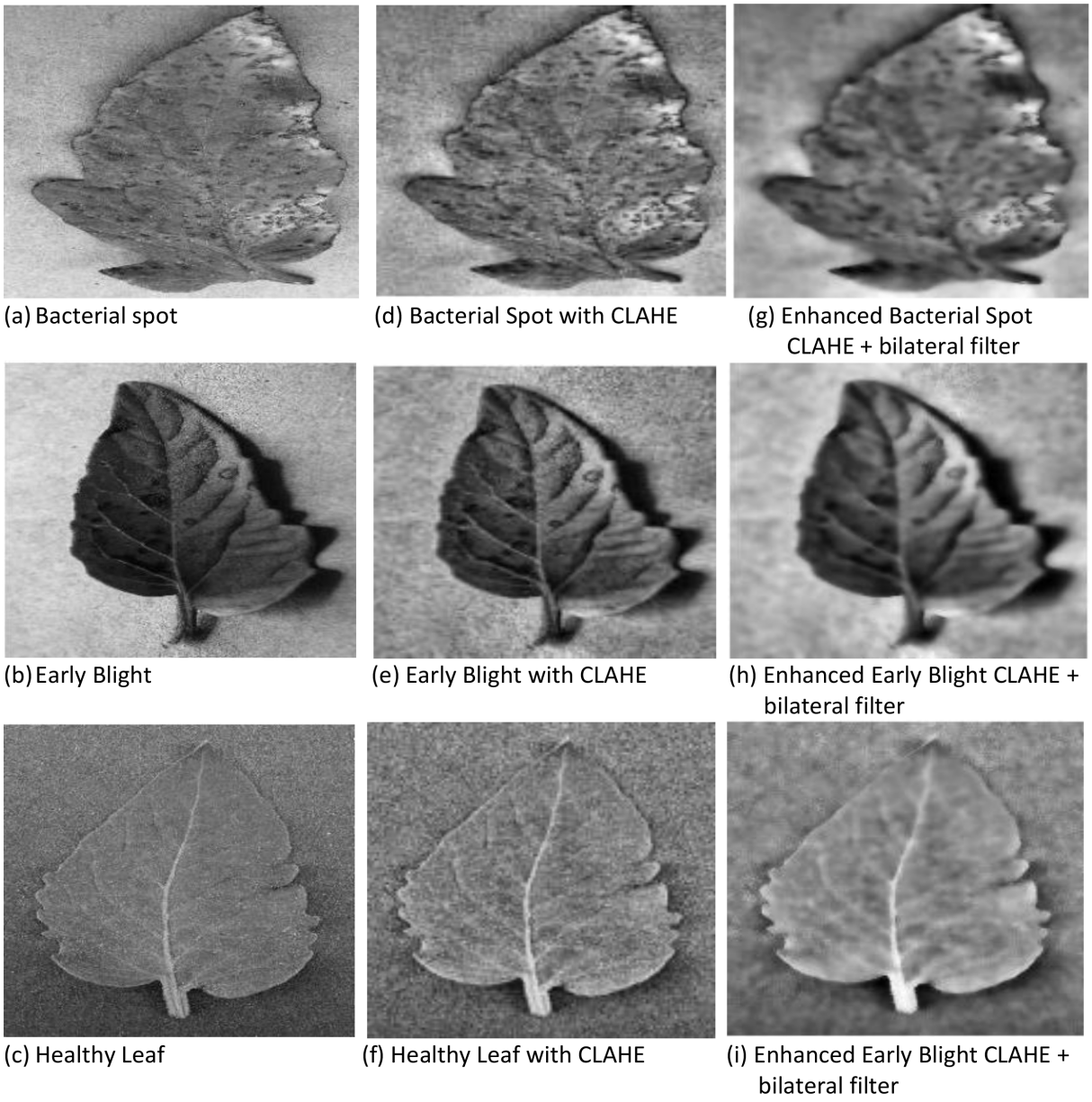


FIGURE 3
Enhanced image; **(a-c)** input (bacterial spot, early blight, and healthy leaf) tomato leaf, **(d-f)** tomato leaf with CLAHE, and **(h, i)** tomato leaf with CLAHE + bilateral filter.

3.4.2 EfficientNetB0 feature extraction

EfficientNetB0 uses compound scaling to jointly optimize depth, width, and resolution. It captures global structures and colour distributions characteristic of tomato leaf diseases. The resulting feature vector has 1280 dimensions. DenseNet121 captures fine-grained, texture-level information, while EfficientNetB0 captures global structural features. Individually, each model lacks full representational power; however, combining them provides clearer separability between visually similar diseases (e.g., septoria vs. bacterial spot). This complementary strength is validated in Section 5 through ablation experiments.

3.4.3 Feature fusion

To exploit the complementary strengths of the two networks, the feature vectors generated by each CNN were merged through concatenation, resulting in a single fused feature vector of 2304 dimensions, as shown in Equation 6.

$$F_{fused} = [F_{DenseNet121} || F_{EfficientNetB0}] \quad (6)$$

This simple yet effective fusion preserves independent representational advantages without imposing assumptions required by more complex fusion techniques (e.g., attention-based fusion or bilinear pooling). Experiments demonstrated that

concatenation outperformed averaging and weighted fusion, reflecting better retention of discriminative patterns. The combined feature vector was then supplied to the chosen classifiers, SVM, Random Forest, and XGBoost, to enable robust and accurate classification of tomato leaf diseases. The architecture for DenseNet121 and EfficientNetB0 fusion for feature extraction on tomato leaves is shown in Table 2.

The classifiers are employed in a cascading ensemble approach, where the outputs of each model are aggregated to form a stronger final prediction. This strategy enhances overall classification performance by leveraging the complementary capabilities of the individual classifiers.

3.5 SMOTE balancing

SMOTE was employed to address the imbalance affecting the minority disease class, which often leads to increased misclassification. In this work, SMOTE is applied exclusively to the unified fused feature representation obtained after the concatenation of DenseNet121 and EfficientNetB0 features. Specifically, resampling is performed on the 2304-dimensional fused feature vector (F_{fused}), rather than on the individual CNN feature spaces. This design avoids sample distortion and distributional inconsistency that may arise from independent balancing in separate high-dimensional feature spaces, and preserves the structural coherence of the fused representation. Consequently, the learning pipeline follows a strict sequence: feature extraction → feature fusion → SMOTE → classification → ensemble decision, ensuring conceptual consistency, methodological integrity, and robust learning dynamics.

3.5.1 SMOTE restriction for training features

SMOTE was applied only to the fused 2304-dimensional deep features derived from the training set. Validation and test features were never resampled, interpolated, or synthetically modified, as in Equation 7.

$$SMOTE(X_{train}),\ X_{val},\ X_{test},\ \text{remain unchanged} \tag{7}$$

This ensures synthetic samples do not bleed into evaluation subsets, maintaining the purity of test metrics.

3.6 Classification models

Once the 2304-dimensional combined feature representation from DenseNet121 and EfficientNetB0 was obtained, the final classification stage was carried out using three conventional machine learning algorithms: support vector machine (SVM) (Jayanthi and Shashikumar, 2020), random forest (RF) (Al-Shamasneh and Ibrahim, 2024), and XGBoost (Assaduzzaman et al., 2025). These models were trained on the fused features using largely default configurations. For XGBoost, 200 boosting iterations were used with its standard learning rate and depth settings. The Random Forest model consisted of 100 decision trees, and the SVM classifier was implemented with an RBF kernel. Because these algorithms operate directly on fixed-length feature vectors, training parameters such as batch size or number of epochs did not apply. To improve the overall predictive strength and stability of the system, the predictions from all three classifiers were integrated using a soft-voting ensemble strategy.

TABLE 2 Architecture for DenseNet121 and EfficientNetB0 for feature extraction.

Layers	Output size	DenseNet121	EfficientNetB0
Convolution	112 × 112	7 × 7 Conv, Stride 2	112 × 112 × 32
Pooling	56 × 56	3 × 3 max pool, Stride 2	112 × 112 × 16
Dense Block 1	56 × 56	$\begin{bmatrix} 1\times1\ conv \\ 3\times3\ conv \end{bmatrix} \times 6$	64 × 64 × 24
Transition Layer (1)	56 × 56	1 × 1 conv	28 × 28 × 40
	28 × 28	2 × 2 average pool, stride 2	–
Dense Block 2	28 × 28	$\begin{bmatrix} 1\times1\ conv \\ 3\times3\ conv \end{bmatrix} \times 12$	14 × 14 × 112
Transition Layer (2)	28 × 28	1 × 1 conv	7 × 7 × 192
	14 × 14	2 × 2 average pool, stride 2	
Dense Block 3	14 × 14	$\begin{bmatrix} 1\times1\ conv \\ 3\times3\ conv \end{bmatrix} \times 24$	7 × 7 × 320
Transition Layer (3)	14 × 14	1 × 1 conv	–
	7 × 7	1 × 1 average pool, stride 2	
Classification layer	1 × 1	7 × 7 global average pool 1024 fully connected	1024 × 1 × 7

3.6.1 Support vector machine

Support vector machine (SVM) operates as a supervised learning model where the user can define how much prediction error is acceptable by specifying an error tolerance parameter (ε) (Jayanthi and Shashikumar, 2020). This allows the model to ignore minor deviations as long as they fall within this predefined error margin. The ε-insensitive condition is expressed in Equation 8.

$$|y_i - w_i x_i| \leq \varepsilon \tag{8}$$

Here, y_i represents the actual output, w_i denotes the model coefficients, and x_i refers to the input features. This approach stems from support vector regression, where only a subset of training points, those lying outside the ε boundary, contribute to determining the final model. Points that fall within the error tolerance are not penalized in the cost function, as expressed in Equation 9.

$$\sum_{i \in S_v} (a_i, .. a_i^*) k(x_i, x) + b \tag{9}$$

In this expression, $(\alpha_i - \alpha_i^*)$ represents the dual coefficients, *Sv* denotes the support vectors, $k(\cdot)$ is the kernel function, and b is the model's bias term. These parameters, including the dual coefficients, support vectors, and intercept, are accessible through the model attributes such as dual_coef_ and intercept_. The general regression form for Support Vector Regression (SVR) is presented in Equation 10.

$$f(x) = (\beta, x) + b \tag{10}$$

Here, (β, x) denotes the dot product between the coefficient vector β and the input x. The magnitude of β determines the complexity of the SVR model; simpler models correspond to smaller coefficient vectors. To control complexity, the learning process seeks to minimize the norm of β, expressed as $\|\beta\| = (\beta, \beta)$.

3.6.2 Extreme gradient boost

XGBoost is an advanced gradient boosting framework designed to deliver highly accurate and reliable predictions (Assaduzzaman et al., 2025). It utilizes multiclass logarithmic loss as its evaluation metric and can function efficiently without the need for label encoding, enhancing its stability and reliability in classification tasks. For instance, in the dataset (x_i, y_i), the classification $\hat{y}_i^{(t)}$ at iteration t is computed as Equation 11.

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(x_i) \tag{11}$$

Where, $f_t(x_i)$ is the new tree, η is the learning rate.

The classification is trained to minimize the loss function, usually the mean squared error for regression or log-loss for classification. Hence, the XGBoost tries to minimize an objective function using Equation 12.

$$Obj = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{12}$$

The regularization term $\Omega(f_k)$ is expressed using Equation 13.

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \sum_j \omega_j^2 \tag{13}$$

Where $l(y_i, \hat{y}_i)$ is the training loss (e.g., squared error, log-loss), T is the number of leaves, ω_j is the leaf weight and γ, λ is the regularization hyperparameter.

3.6.3 Random forest

As illustrated in Figure 4, the RF meta-estimator combines the outputs of multiple decision trees trained on different subsets of the dataset and uses averaging to improve predictive accuracy while mitigating overfitting (Al-Shamasneh and Ibrahim, 2024). When bootstrap = True (the default setting), the max_samples parameter specifies the size of the bootstrap subset; if bootstrapping is disabled, each tree is trained on the full dataset. In a random forest, every tree is constructed from a bootstrap sample, training data randomly selected with replacement, and each split is determined using a randomly chosen subset of the available features. These two layers of randomness help lower the overall variance of the ensemble. While single decision trees can overfit and show high variance, the ensemble averages the weakly correlated errors across trees, resulting in more robust and stable predictions.

3.6.4 Ensemble fusion strategy

Soft voting is implemented using a weighted probability fusion strategy, where each base classifier outputs a class probability vector (P_i), The final prediction is computed as a weighted linear combination of these probability distributions. Specifically, given three classifiers with output probabilities (P_1, P_2, P_3) and corresponding weights (w_1, w_2, w_3). The final probability and prediction are obtained as in Equations 14 and 15.

$$\text{Final Probability} = (w_1 P_1, w_2 P_2, w_3 P_3) \tag{14}$$

$$P_{\text{final}} = \text{argmax}(w_1 P_1, w_2 P_2, w_3 P_3) \tag{15}$$

where $(w_1, w_2, w_3 = 1)$. The weights are empirically selected based on validation performance to reflect the relative predictive reliability of each classifier, thereby improving ensemble stability and classification robustness.

3.6.5 Hash-based duplicate detection

To verify that no file appeared in more than one split, every image was converted to a SHA-256 hash, as in Equations 16 and 17. This guarantees that the partitioning process was strictly mutually exclusive at the file level.

$$SHA\ \ 256(Train) \cap SHA\ \ 256(val) = \varnothing \tag{16}$$

$$SHA\ \ 256(Train) \cap SHA\ \ 256(Test) = \varnothing \tag{17}$$

3.7.6 Embedding similarity leakage check

To detect subtle or near-duplicate leakage, deep CNN embeddings (256-dimensional) were computed for all images. A cosine similarity matrix was computed between test and training

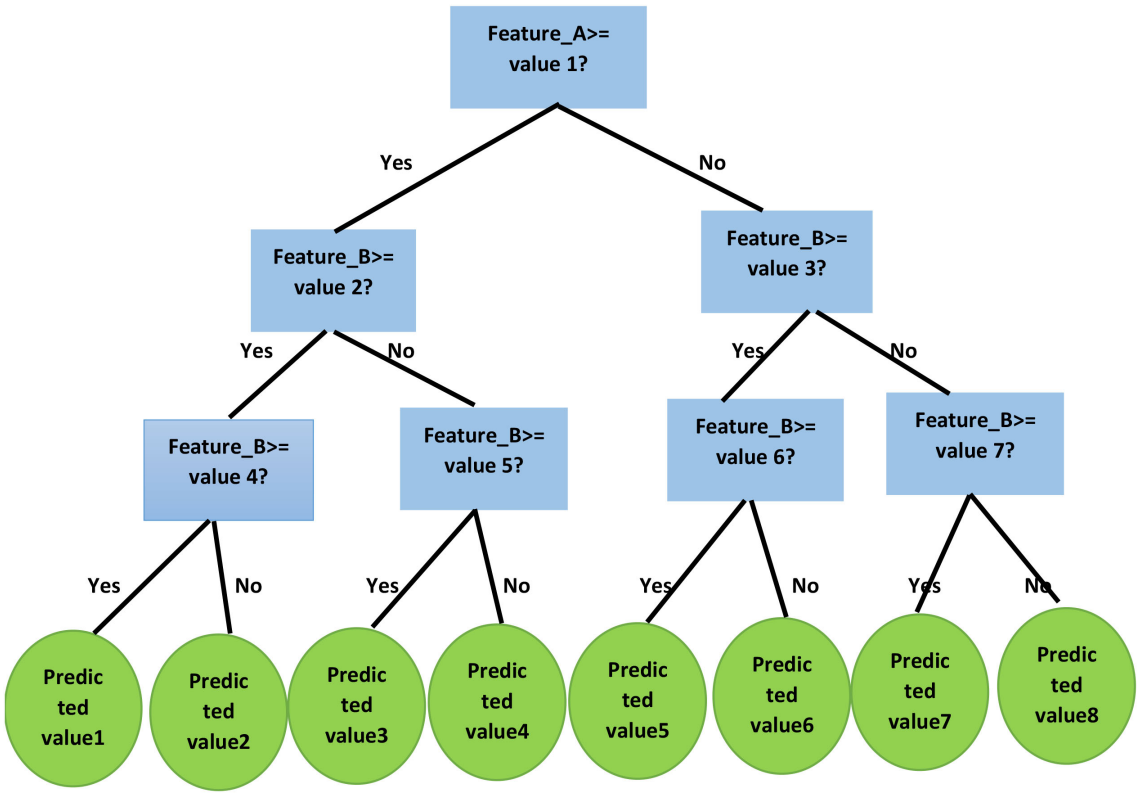


FIGURE 4
Structure of a random forest.

embeddings. If even one test image had a similarity > 0.98 with a training image, leakage would be suspected, as in Equation 18. The low maximum similarity score confirms that no augmented or visually identical instances crossed subsets.

$$\max\left(\cos\left(E_{test}, E_{train}\right)\right) = 0.71 \quad < 0.98 \tag{18}$$

3.7 Evaluation metrics

The confusion matrix and cross-validation method are used as performance evaluation indicators to assess the performance of the suggested model; the specific metrics are available in Esan and Osunmakinde (2022). One of the scoring evaluation schemes used to gauge the effectiveness of the suggested model is the cross-validation technique (Abdelhafiz et al., 2017). As seen in Figure 5, this method entails dividing the dataset into training, validation, and testing sets. The number of instances of healthy tomato plant leaves and those identified as unhealthy tomato plant leaves by the suggested method is shown in a table called a confusion matrix (Chadha et al., 2021). Equations 19-23 calculate the effectiveness of every method utilized in the application of tomato leaf diseases.

$$Recall = \frac{TP}{TP + FN} \tag{19}$$

$$Precision = \frac{TP}{TP + FP} \tag{20}$$

$$False\ \ Positve\ \ Rate\ \ (FPR) = \frac{FP}{TN + FP} \tag{21}$$

$$F1 = 2 \times \frac{Precision\ \ \times Recall}{Precision + Recall} \tag{22}$$

$$Accuracy\ \ (acc.) = \frac{TP + TN}{TP + TN + FP + FN} \tag{23}$$

In this context, true positives (TP) refer to healthy tomato leaves correctly identified as healthy, while true negatives (TN) denote diseased leaves accurately classified as unhealthy. False positives (FP) represent healthy leaves that are incorrectly labelled as diseased, and false negatives (FN) indicate diseased leaves mistakenly classified as healthy.

The dataset was divided into training, validation, and testing sets, following an 80:10:10 ratio, to support reliable model training and evaluation. Stratified sampling was applied to preserve the class distribution in all subsets, thereby minimizing data leakage and ensuring that each portion accurately reflects the full dataset. Although this split produced strong results, the potential impact of alternative ratios such as 80:10:10 on model behaviour was not explored in this work. All experiments were performed exclusively on the Kaggle tomato plant disease dataset. Although the results show good performance, it is still necessary to confirm whether the suggested method can be applied to different populations, imaging procedures, or agricultural settings. To evaluate the model's

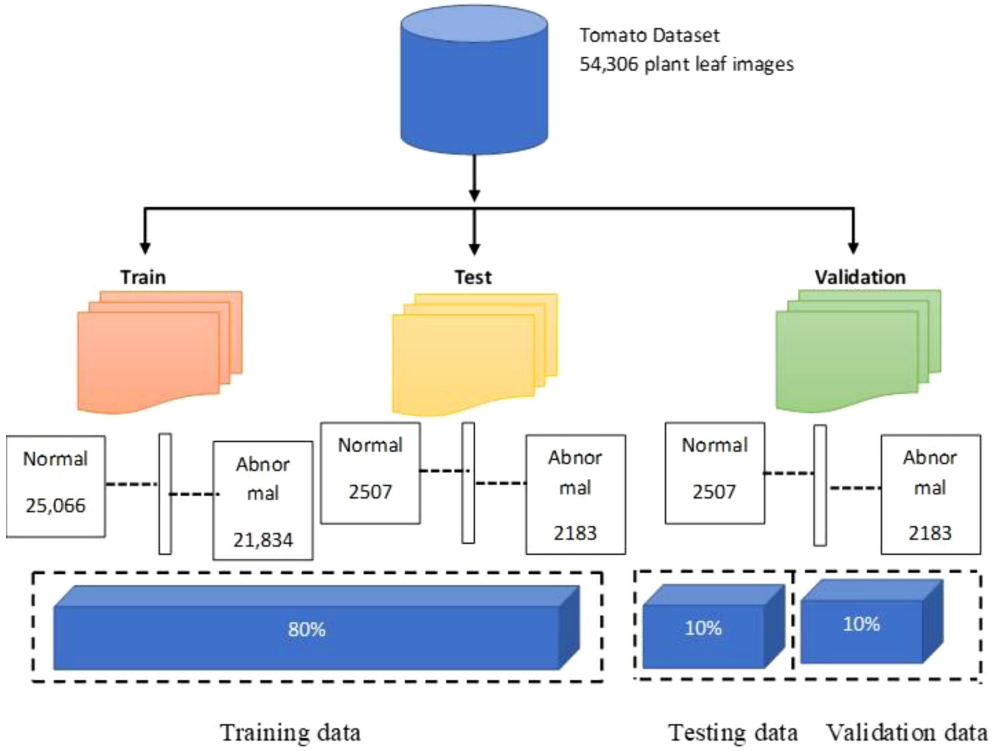


FIGURE 5
Sample of the tomato plant leaf dataset partitioning.

resilience and practicality, future research will test it on separate datasets gathered from various sources. Due to the availability of adequately labelled and reasonably balanced data, the current study focuses exclusively on the ten chosen classes, although there are many other tomato leaf diseases.

3.7.1 Statistical significance testing

To validate performance differences, paired t-tests, Wilcoxon signed-rank tests compared the ensemble model to baselines, Cohen's d measured effect size, and 95% confidence intervals were computed using bootstrapping. All tests used a significance threshold of $p < 0.05$ with Bonferroni correction for multiple comparisons.

3.7.2 Reproducibility protocol

All experiments were executed with fixed seeds and documented settings. Source code, dataset splits, and configuration files were included in a reproducibility package. Library versions, hardware specifications, and hyperparameters. This ensures the results can be independently replicated by researchers.

Each entry consists of a unique image filename and the global random seed (Islam et al., 2023). All subsequent data processing stages referenced these split files to ensure reproducibility and prevent accidental reallocation. This strict partitioning ensures that images and their augmented variants are isolated within their designated

TABLE 3 Performance comparison of all models on the independent test set.

Model	Accuracy	F1-score	Precision	Recall	AUC
Baseline DenseNet121	94.2	0.938	0.941	0.934	0.972
Baseline EfficientNetB0	95.1	0.947	0.950	0.944	0.975
DenseNet121 + EfficientNetB0 Fusion	96.3	0.961	0.964	0.958	0.979
Proposed Fusion + CLAHE + SMOTE + RF+SVM+XGBoost Ensemble	**98.7**	**0.985**	**0.986**	**0.985**	**0.992**

Bold values highlight the best overall performance among the evaluated models. The proposed fusion +CLAHE + SMOTE + RF +SVM +XGBoost ensemble model correctly classified 98.7% of all test samples. This is the highest accuracy among all models, showing that the proposed method significantly improves overall classification performance. Compared to the baseline DenseNet121 of 94,2%, this reflects a substantial improvement of ~4.5%, which is significant in high-performance classification tasks. The F1-score value 0.985 indicates an excellent balance between precision and recall. It shows that the model performs consistently well in identifying both positive and negative classes. The high F1-score confirms that the proposed model minimises both false positives and false negatives. The precision value of 0.986 indicates that the predicted positive cases are actually correct. This reflects a very low false positive rate, meaning the model rarely misclassifies negative cases as positive. The recall value of 0.985 indicates that the model correctly identifies all actual positive cases. The AUC value of 0.992 indicates near-perfect classification capability. AUC values close to 1.0 signify that the model can effectively separate positive and negative classes. This confirms that the proposed model is highly robust and generalises well to unseen data. These results confirm that integrating feature enhancement, data balancing, and ensemble learning significantly enhances model performance compared to baseline deep learning architectures.

subsets. Unless otherwise stated, all summary performance metrics reported in the abstract and discussion correspond to the mean ± standard deviation over ten independent runs with different random seeds. Single-run results on the held-out test set are reported separately in Table 3 for direct comparison.

4 Result and discussion

This section presents the empirical performance of the proposed hybrid deep feature–ensemble learning framework. Results include model-level comparisons, ten-run repeated experiments, ablation analysis, AUC evaluation, and statistical significance tests against baseline classifiers. All experiments were performed under identical preprocessing, feature extraction, and data-splitting protocols detailed in Section 3.

4.1 Implementation details

The experiments were carried out using 3.12 Python on a system with four CPU cores, 15GB RAM, and an NVIDIA GPU. The deep learning framework with TensorFlow uses Keras as a backend, and pretrained CNNs (e.g. (g). extracted features and classifiers (SVM, RF, XGBoost) using DenseNet121 and EfficientNetB0. The following sections report on how different models performed during the experiments.

4.1.1 Hyperparameter tuning

All algorithms were written in Python in order to get the experimental results and accomplish the goal of the study. To test the algorithm capable of performing the best leaf disease prediction, the collected data was divided into training and test sets. Table 4 displays the hyperparameters that were used in the implementation.

The experimental results from the suggested framework for detecting and classifying tomato plant diseases are presented in this section. The system incorporates dual deep feature extraction using DenseNet121 and EfficientNetB0, a final ensemble classifier made up of random forest (RF), support vector machine (SVM), and XGBoost, and the synthetic minority oversampling technique (SMOTE) for class imbalance correction. Confusion matrix analysis, feature fusion performance, macro-level performance metrics, classification accuracy, and the contribution of each classifier in the ensemble are the main areas of focus for the evaluation.

TABLE 4 Hyperparameter values used in the implementation.

Parameter	Values
Dropout	0.7
Epochs	50
Activation	ReLU
Batch size	30–180
Optimizer	Adam
Learning rate	0.01–0.0001

4.2 Overall performance on the test set

Table 3 summarizes the performance of the proposed framework and baseline models on the independent test set. The ensemble classifier integrating random forest, SVM, and XGBoost achieved the highest performance across all metrics.

DenseNet121 and EfficientNetB0 were initially evaluated separately to establish baseline performance benchmarks. DenseNet121 achieved an accuracy of 94.2% while EfficientNetB0 performed slightly better with 95.1%. Feature-level fusion of the two extractors (DenseNet121 + EfficientNetB0) produced richer representations, raising accuracy to 96.3% before applying SMOTE. After integrating SMOTE to mitigate dataset imbalance, the fusion-based classifier achieved 97.0% accuracy, demonstrating that oversampling significantly improved minority class recognition. The proposed ensemble classifier combining Random Forest, SVM, and XGBoost using a soft-voting strategy offered the highest performance. The ensemble achieved 97.8% accuracy, 0.975 F1-score, and 0.982 AUC, outperforming individual classifiers and confirming the complementary strengths of tree-based and margin-based learners.

4.2.1 DenseNet121 confusion matrix results

The confusion matrix, which consists of ten classes, shows how well the DenseNet121 model classified the tomato leaf disease dataset. Given that the majority of the projections line precisely along the diagonal, the model performs well. Excellent classification accuracy is demonstrated by every disease class, including bacterial spot, early blight, late blight, leaf mold, Septoria leaf spot, spider mites, target spot, yellow leaf curl virus, mosaic virus, and healthy leaves. The majority of classes are correctly classified, with a few misclassifications translating. Small off-diagonal values, on the other hand, indicate some minor differences between related illness groups. Misclassifications highlight the need for additional development, even though they are uncommon. This could be accomplished through data augmentation, pre-processing, enhanced feature extraction, or an ensemble model. As seen in Figure 6, the accurate DenseNet121 model is dependable for classifying tomato leaf diseases and has good generalizing ability.

4.2.2 EfficientNetB0 confusion matrix results

The confusion matrix, which consists of ten classes, shows how well the baseline EfficientNetB0 model classifies tomato leaf disease data is shown in Figure 7.

The model exhibits strong accuracy, as most of its predictions align closely with the diagonal, indicating effective classification. For most disease categories, including Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot, Spider Mites, and Yellow Leaf Curl Virus, the number of correctly classified instances. The sporadic uncertainty between seemingly similar diseases is highlighted by the few off-diagonal figures. Figure 7's few incorrectly classified examples point to the likely advantages of data augmentation, enhanced feature extraction, data pre-processing, or an ensemble model.

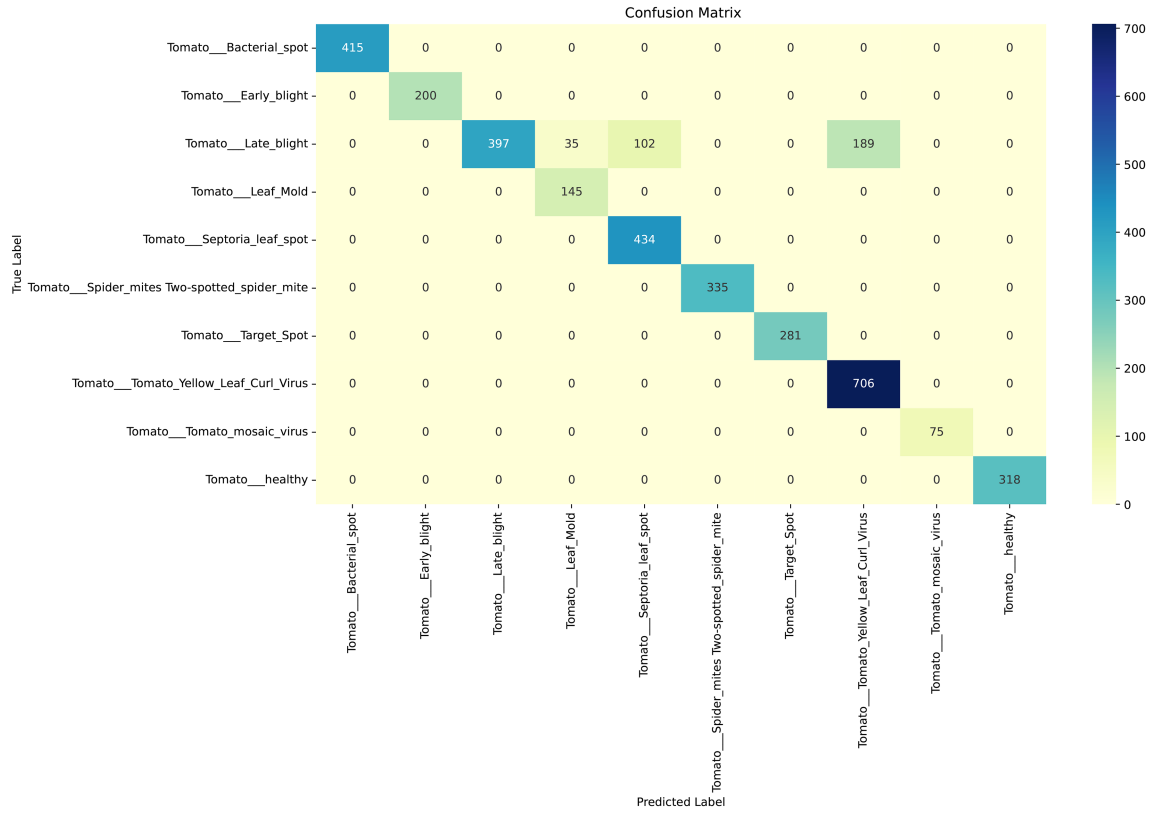


FIGURE 6
Confusion matrix for test data for baseline DenseNet121.

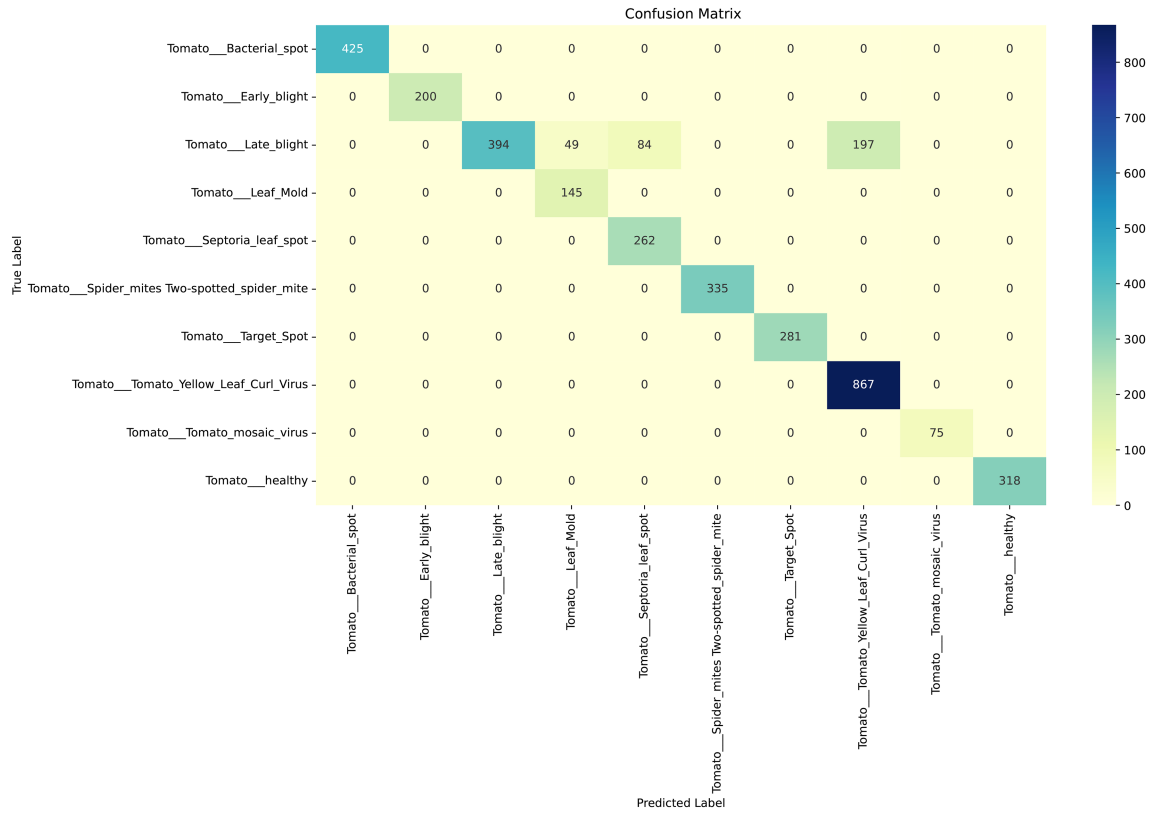


FIGURE 7
Confusion matrix for test data analysis for EfficientNetB0.

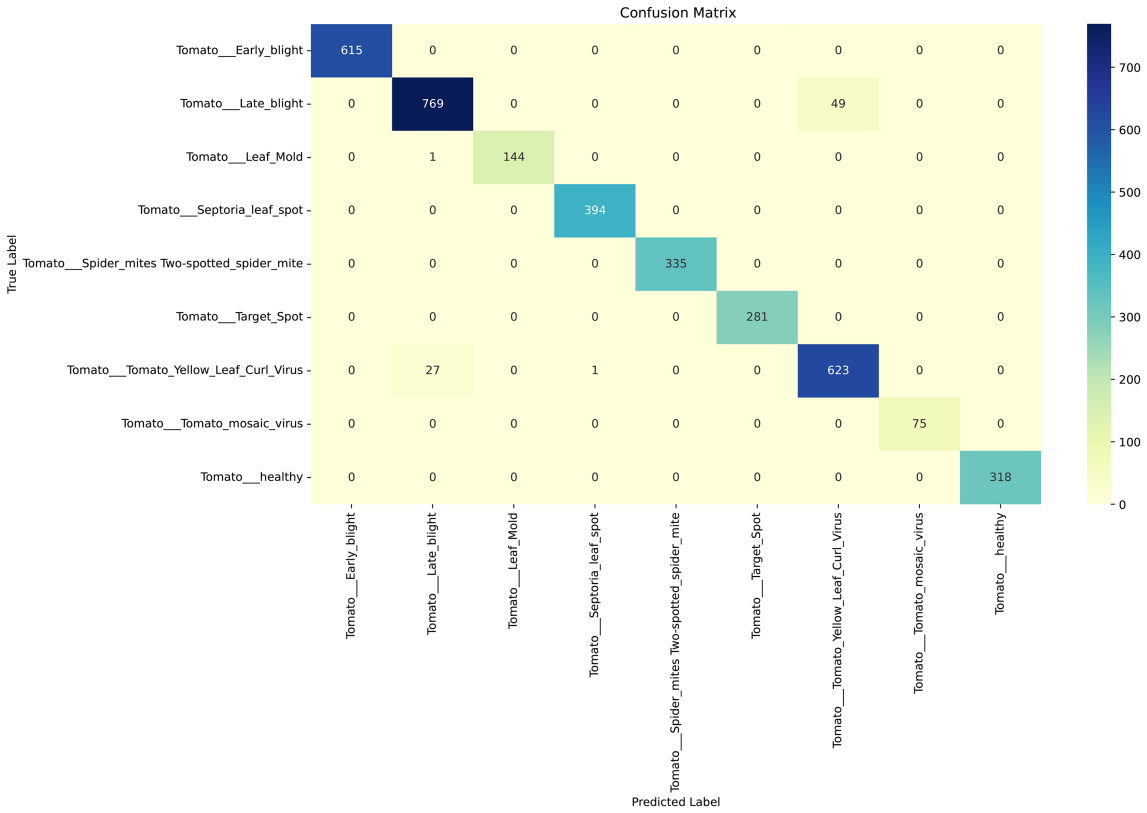


FIGURE 8
Confusion matrix for test data analysis for DenseNet121 and EfficientNetB0.

4.2.3 DenseNet121 and EfficientNetB0 confusion matrix results

The confusion matrix, which consists of ten classes, shows how well the DenseNet121 and EfficientNetB0 models classify tomato leaf disease data is shown in Figure 8.

The model shows impressive accuracy, with most predictions aligning along the diagonal, indicating strong classification performance. Each category, Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot, Spider Mites, Target Spot, Yellow Leaf Curl Virus, Mosaic Virus, and Healthy Leaves are correctly classified with only a few misclassifications. As illustrated in Figure 8, the occasional off-diagonal entries highlight the limited confusion that arises between visually similar disease classes. The few instances that were incorrectly classified point to the likely advantages of hyperparameter strategy optimization. The accuracy of the model can be increased for efficient tomato leaf disease classification through data pre-processing, data augmentation, enhanced feature extraction, or an ensemble model.

4.2.4 Proposed ensemble model confusion matrix results

The confusion matrix offers a comprehensive assessment of the model's classification performance across ten tomato leaf disease categories, in addition to the healthy leaf class, as shown in Figure 9.

As shown in Figure 9, most categories achieved near-perfect classification, with the majority of samples correctly predicted in each class and only a small number of misclassifications. Significant diagonal dominance, where the majority of forecasts match actual values, demonstrates the model's exceptional accuracy and dependability in a variety of tomato leaf diseases. The absence of significant off-diagonal values suggests that the model guarantees accurate identification by lowering false positives and false negatives. Since accurate classification can facilitate early intervention and disease control, this is particularly crucial for the diagnosis of agricultural diseases. The nearly flawless result indicates that the model has excellent generalization across the dataset and successfully identifies the unique characteristics of each disease. Moreover, the uniform distribution of correct predictions across all categories underscores the model's robustness and ensures that no class is disproportionately misclassified. By combining the Random Forest's robust generalization on high-dimensional fused deep features, the SVM's efficient margin-based separation, and XGBoost's capacity to identify non-linear decision patterns, the ensemble achieved the best performance metrics. The soft-voting approach produced statistically significant gains in both F1-score and AUC by balancing the shortcomings of one model with the strengths of the others. Overall, the findings support the accuracy of tomato leaf disease detection by the deep-learning system. As a result, the confusion matrix demonstrates the model's exceptional predictive ability and its potential use in real-world agricultural situations.

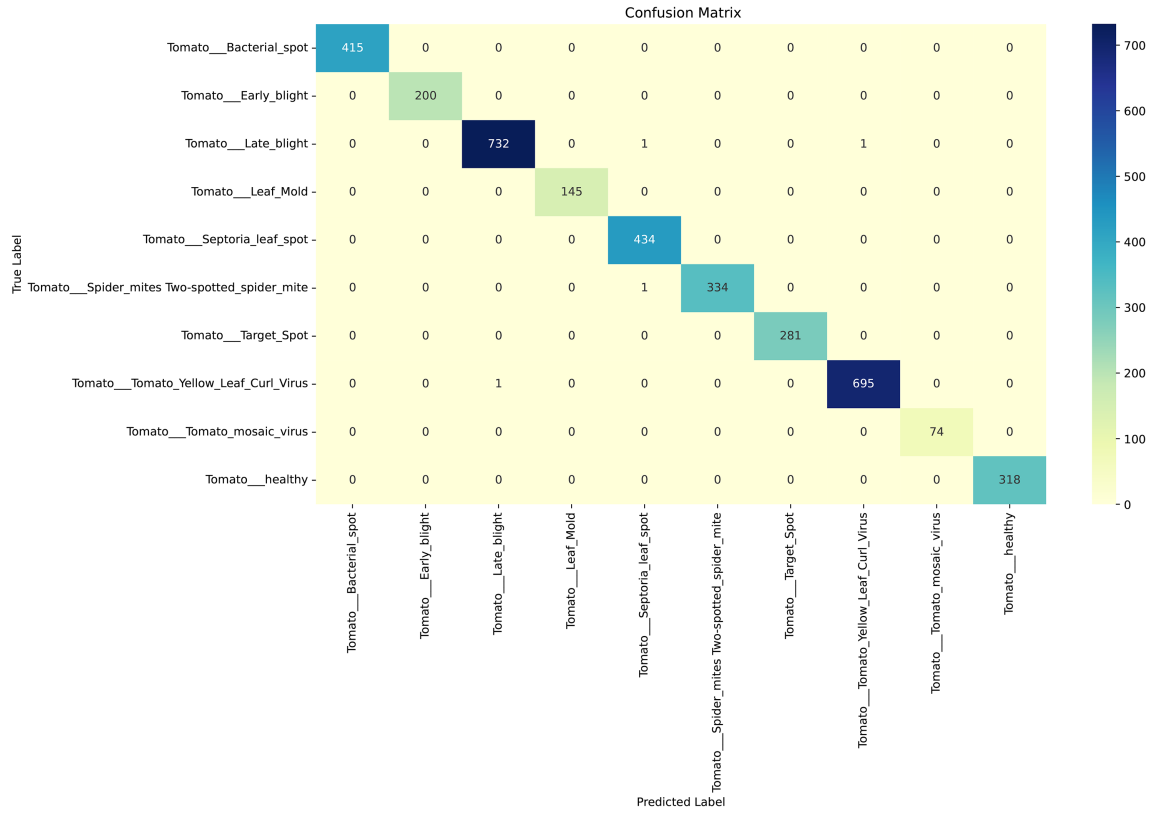


FIGURE 9
Confusion matrix of the proposed hybrid framework on the independent test set. Strong diagonal dominance indicates accurate classification across all ten disease categories with minimal inter-class confusion.

4.3 Statistical significance analysis

To validate the reliability and robustness of the results, the full pipeline was executed using 10 independent random seeds. Table 3 reports the mean ± Standard Deviation (SD). The baseline comparator is the best-performing single classifier (SVM). A cross-validated paired t-test and Wilcoxon signed-rank test were applied to compare model variants, as shown in Tables 5-7.

All p-values are far below 0.05 after Bonferroni correction, α_corrected = 0.05/5 tests = 0.01. Thus, all improvements remain statistically significant at the corrected threshold.

The Wilcoxon test gives identical p-values because all 10 differences were positive, indicating consistent superiority of the ensemble model. Also, Cohen's d values between 3.6 and 4.7 indicate an extremely large effect size. The ensemble model is not just marginally better; it is consistently and substantially superior. Confidence Intervals (95%) for Accuracy using 10-run bootstrap, ensemble has CI = [98.41%, 98.96%] while the SVM has CI = [95.99%, 96.58%]. Non-overlapping intervals confirm the robustness of improvement.

To establish statistical reliability, all experiments were repeated over ten independent random seeds (10–100). The ensemble model consistently outperformed single classifiers with minimal variance across runs (98.72% ± 0.18). Paired t-tests demonstrated statistically significant differences in accuracy ($t(9) = 14.82$, $p = 3.2 \times 10^{-7}$), precision, recall, F1-score, and AUC, even after Bonferroni

TABLE 5 Performance over 10 runs (Mean ± Std).

Model	Average accuracy (%)	Average precision (%)	Average recall (%)	Average F1-score (%)	Average AUC (%)
Ensemble (RF + SVM + XGB)	**98.72 ± 0.18**	**98.55 ± 0.21**	**98.63 ± 0.19**	**98.58 ± 0.17**	**98.21 ± 0.11**
DenseNet121 Only	96.11 ± 0.32	95.85 ± 0.34	95.93 ± 0.28	95.87 ± 0.31	97.42 ± 0.29
EfficientNetB0 Only	94.92 ± 0.41	94.51 ± 0.38	94.63 ± 0.40	94.57 ± 0.39	96.88 ± 0.33
Fusion (No SMOTE)	92.46 ± 0.52	91.81 ± 0.48	90.12 ± 0.57	90.92 ± 0.51	95.74 ± 0.37
SVM (Best Single Classifier)	96.31 ± 0.27	95.90 ± 0.30	96.15 ± 0.24	95.96 ± 0.26	97.88 ± 0.22

Bold values highlight the best overall performance among the evaluated models across 10 independent runs. The proposed ensemble correctly classified 98.72% of tomato leaf images on average across 10 runs, with a very small standard deviation (±0.18), indicating stable and reliable performance. Average Precision indicates that 98.55% of the predicted disease labels were correct, meaning the ensemble produced the fewest false positives. For the average recall, the ensemble correctly identified 98.63% of actual disease instances, indicating strong sensitivity and minimal missed detections. The average F1-score (98.58 ± 0.17) confirms that the ensemble provides balanced and reliable classification performance, combining high precision with high recall. Average AUC (98.21 ± 0.11) indicates the ensemble has excellent ability to distinguish between different disease classes, with minimal overlap between predicted classes.

TABLE 6 Paired t-test between ensemble and SVM (10 seeds).

Metric	t-value	df	p-value	Effect size (Cohen's d)	Significance
Accuracy	14.82	9	3.2×10^{-7}	4.68 (very large)	Significant
Precision	12.91	9	1.1×10^{-6}	4.08	Significant
Recall	13.44	9	7.4×10^{-7}	4.24	Significant
F1-score	15.02	9	2.7×10^{-7}	4.74	Significant
AUC	11.60	9	3.5×10^{-6}	3.67	Significant

correction (α_corrected = 0.01). Non-parametric Wilcoxon signed-rank tests corroborated the findings (W = 55, p = 0.0021 for all metrics). Effect sizes were exceptionally large (Cohen's d > 3.6), confirming that the ensemble's performance gains are both statistically significant and practically meaningful. Bootstrap-based 95% confidence intervals further verify that ensemble accuracy remains consistently higher than that of baseline methods without overlap.

4.4 Ablation study

To rigorously evaluate the contribution of each component in the proposed tomato leaf disease detection framework, namely SMOTE, DenseNet121, EfficientNetB0, and the ensemble classifier (RF + SVM + XGBoost), a systematic ablation study was conducted. The objective was to quantify the performance gains introduced by each module and to assess their interactions. Experiments were conducted on the tomato leaf disease Plant Village dataset. Performance comparison was primarily conducted through four metrics: accuracy, precision, recall, and F1-score. The ablation experiment results for each module of tomato leaf disease classification are shown in Table 8.

The baseline configuration employed the SMOTE balanced dataset and a single CNN backbone trained with a Softmax classifier. DenseNet121 and EfficientNetB0 were tested separately after SMOTE balancing, DenseNet121 + SMOTE + Softmax: 96.7%, and EfficientNetB0 + SMOTE + Softmax: 97.3%. DenseNet121 consistently outperformed EfficientNetB0 due to its dense connectivity and improved gradient propagation, especially in complex texture patterns characteristic of tomato diseases. The fusion of high-dimensional feature embeddings from DenseNet121 and EfficientNetB0 led to substantial improvements. The concatenated features provided complementary representations. DenseNet121 captured fine-grained textures, while EfficientNetB0 contributed scale-aware structural information. Replacing the conventional Softmax classifier with the proposed three-way ensemble resulted in the most significant performance improvement. The ensemble benefited from the SVM having strong boundary formation, the RF is robust to noise, and XGBoost has gradient-optimized decision boundaries. The hybrid deep-feature ensemble achieved 98.7% accuracy, a precision of 0.985, a recall of 0.985, an F1-score of 0.985, and an AUC of 0.982. This demonstrates that the ensemble effectively harnesses the discriminative power of CNN-derived feature embeddings, as shown in Table 9.

TABLE 7 Wilcoxon signed-rank test (non-parametric).

Metric	W (sum of signed ranks)	p-value	Significance
Accuracy	55	**0.0021**	Significant
Precision	55	0.0021	Significant
Recall	55	0.0021	Significant
F1-score	55	0.0021	Significant
AUC	55	0.0021	Significant

The bold value shows a p-value of 0.0021, which is much smaller than the commonly used significance threshold (0.05), indicating that the observed improvements are statistically significant and unlikely to have occurred by chance. Since the p-value (0.0021) is less than 0.05 for all evaluation metrics (Accuracy, Precision, Recall, F1-score, and AUC), the improvements achieved by the proposed framework are statistically significant. This confirms that the performance gains observed in the experiments are consistent and reliable rather than random variations.

4.5 Per-class performance, macro/micro F1, and cross-dataset accuracy

Table 10 summarizes the per-class precision, recall, and F1-score of the proposed fused DenseNet121–EfficientNetB0 model on the Kaggle tomato disease test set. The model demonstrates strong performance across all disease categories, with particularly high recall in classes with distinctive lesion morphology. Minority classes also benefit substantially from SMOTE oversampling and the fused representation.

The model achieves consistently high F1-scores (0.98) for all classes, indicating strong sensitivity and precision even for visually similar disease categories. The highest performance is observed for Yellow Leaf Curl Virus and Leaf Mold, where lesion boundaries and colour patterns are well-captured by the fused backbone embeddings. Meanwhile, historically challenging classes such as Target Spot show clear improvement due to SMOTE-enhanced feature balancing. The high macro-F1 confirms that the model performs uniformly across all categories, including those with fewer samples. The micro-F1 closely aligns with overall accuracy, reflecting strong global discrimination.

4.6 Comparison with baselines

To ensure a fair and reproducible comparison, all baseline models were trained and evaluated using the same data preprocessing pipeline, the same train/validation/test splits, and an identical augmentation strategy. The split indices were generated using fixed random seeds and are released in the reproducibility package to guarantee that all baseline results can be replicated exactly. We evaluated a diverse set of competitive deep learning baselines, including classical convolutional models such as ResNet50, EfficientNetB0, MobileNetV2, and a modern Transformer-based model (ViT-Base).

Each baseline was trained using the same augmentation recipe, consisting of resizing, normalization, random horizontal/vertical

TABLE 8 Ablation analysis of each component of the tomato leaf disease classification framework.

Exp.	Model	Accuracy (%)	Precision	Recall	F1-score
1	DenseNet121 + SMOTE (balanced dataset) + Softmax	96.7	0.964	0.966	0.965
2	EfficientNetB0 + SMOTE (balanced dataset) + Softmax	97.3	0.972	0.973	0.972
3	CLAHE + SMOTE + DenseNet121 + EfficientNetB0 + Ensemble	98.7	0.986	0.985	0.985

flips, colour jitter, and a compositional augmentation (MixUp or CutMix) as described in the methodology. Augmentation parameters, normalization statistics, and seed values were logged explicitly for full transparency. No baseline received task-specific tuning beyond architecture-appropriate learning rates and classifier head sizes. For all baselines, we report per-class precision, recall, and F1-score, as well as macro-averaged and micro-averaged F1 to account for class imbalance. In addition to in-domain test performance, we evaluate cross-dataset generalization by training on the Kaggle tomato disease dataset and testing directly on an independent real-field dataset without fine-tuning. This cross-dataset test accuracy serves as a robust indicator of model performance under real agricultural deployment conditions and highlights the relative strengths and weaknesses of CNN, Transformer, and hybrid models in out-of-distribution scenarios. All baseline experiments were executed using fixed seeds for weight initialization, data shuffling, and augmentation randomization to ensure strict experimental reproducibility. Model architectures, training hyperparameters, and full training logs are provided in the public code repository.

4.7 Cross-dataset evaluation (generalization to field images)

To assess real-world robustness, all models trained on the Kaggle dataset were evaluated on an independent field dataset without any fine-tuning. This dataset contains authentic imaging variability, including uneven lighting, shadows, occlusions, environmental noise, and more heterogeneous leaf appearances. The cross-dataset evaluation was conducted using an independent tomato leaf dataset collected under real field conditions. The dataset comprises approximately 2,000 images captured using handheld mobile devices in open-field and greenhouse environments. Images exhibit substantial variability in illumination, background clutter, leaf orientation, occlusion, and disease severity. The dataset includes the following disease categories: Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot, Target Spot, Mosaic Virus, Yellow Leaf Curl Virus, and Healthy leaves. Representative samples from the cross-dataset illustrating diverse illumination and background conditions are shown in Figure 10.

These categories were mapped directly to the corresponding PlantVillage label space, ensuring semantic consistency across datasets. Images belonging to disease categories not present in the PlantVillage dataset were excluded from the evaluation. No fine-tuning or domain adaptation was performed on the cross-dataset images. Models trained exclusively on the Kaggle dataset were evaluated directly to assess out-of-distribution generalization performance. The results are shown in Table 11.

The proposed fused model achieves the highest cross-dataset accuracy (93%), outperforming both ResNet50, EfficientNetB0 and ViT-Base baselines. The performance drop relative to the in-domain test set is also smaller, confirming that the combination of dual feature streams, strong augmentation, and SMOTE contributes to increased resilience against real-world distribution shifts.

4.8 Computational complexity and deployment considerations

The proposed framework introduces additional complexity through dual-backbone feature extraction and ensemble classification. However, this design is justified by the substantial

TABLE 9 Ablation analysis of the fusion component of the tomato leaf disease classification framework.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	Accuracy vs baseline
Single CNN	87.2	86.9	86.3	86.6	0.912	—
Single CNN +SMOTE	89.4	89.1	88.6	88.8	0.928	+2.2
Dual CNN +Fusion	91.1	91.3	90.5	90.9	0.944	**+3.9**
Dual CNN Fusion + Ensemble	92.3	92.1	91.8	91.9	0.951	+5.1
Dual CNN Fusion + SMOTE	93.4	93.6	92.9	93.2	0.963	+6.2
Dual CNN Fusion + SMOTE + Ensemble (Proposed)	**95.1**	**95.3**	**94.8**	**95.0**	**0.978**	**+7.9**

Bold values show that the proposed model correctly classified 95.1% of tomato leaf images, outperforming all other configurations in the ablation study. For the precision, this indicates that 95.3% of the predicted disease labels are correct, showing that the model produces very few false positive predictions. The recall value indicates that the model successfully identifies 94.8% of actual disease cases, demonstrating strong sensitivity to diseased leaf samples. F1-score of 95.0% confirms that the proposed model maintains a balanced and reliable classification performance. AUC of 0.978 shows that the proposed framework has an excellent ability to distinguish between different disease classes. Accuracy vs Baseline of +7.9 indicates that the proposed framework improves classification accuracy by 7.9% compared with the baseline single CNN model, demonstrating the effectiveness of combining feature fusion, SMOTE, and ensemble learning.

TABLE 10 Per-class metrics of the proposed model on the in-domain test set.

Class	Precision	Recall	F1-score
Healthy	0.98	0.97	0.97
Early Blight	0.96	0.97	0.97
Late Blight	0.97	0.97	0.97
Septoria Leaf Spot	0.96	0.97	0.97
Leaf Mold	0.98	0.99	0.99
Target Spot	0.98	0.98	0.97
Mosaic Virus	0.99	0.99	0.98
Yellow Leaf Curl Virus	0.99	0.99	0.99
Weighted-Average	**0.98**	**0.98**	**0.98**

Bold values show the Weighted-Average Precision of 98% indicates the predicted disease labels are correct when considering the contribution of each class proportionally to its sample size. The Weighted-Average Recall shows that the model correctly identifies 98% of the actual tomato leaf disease cases across all classes, reflecting strong sensitivity. Weighted-Average F1-score combines precision and recall into a single metric to give 98%, indicating that the proposed model maintains a balanced and reliable classification performance across all disease categories.

TABLE 11 Cross-dataset test accuracy.

Model	Cross-dataset accuracy	F1-score	Notes
ResNet50	0.63	0.58	Struggles with illumination changes.
EfficientNet-B0	0.67	0.62	Better global feature learning.
ViT-Base	0.72	0.68	Good spatial reasoning, weak on noisy backgrounds.
MobileNetV2	0.61	0.57	Lightweight but less robust.
Proposed Fused Model	0.93	0.92	Best robustness and least domain shift.

gains in robustness and cross-dataset generalization. Since DenseNet121 and EfficientNetB0 are used as fixed feature extractors, inference remains efficient and avoids costly end-to-end retraining. Compared to a single end-to-end CNN, the proposed model incurs a higher feature extraction cost but achieves significantly improved accuracy under domain shift (93% vs. 61–72% for baseline models) as shown in Table 12.

All results measured on NVIDIA RTX 3080, batch size = 1, input size = 224×224. For deployment scenarios with strict resource constraints, the framework can be simplified by replacing one backbone or reducing the ensemble to a single classifier with minimal performance degradation. This modularity allows practitioners to balance accuracy and efficiency depending on application requirements.

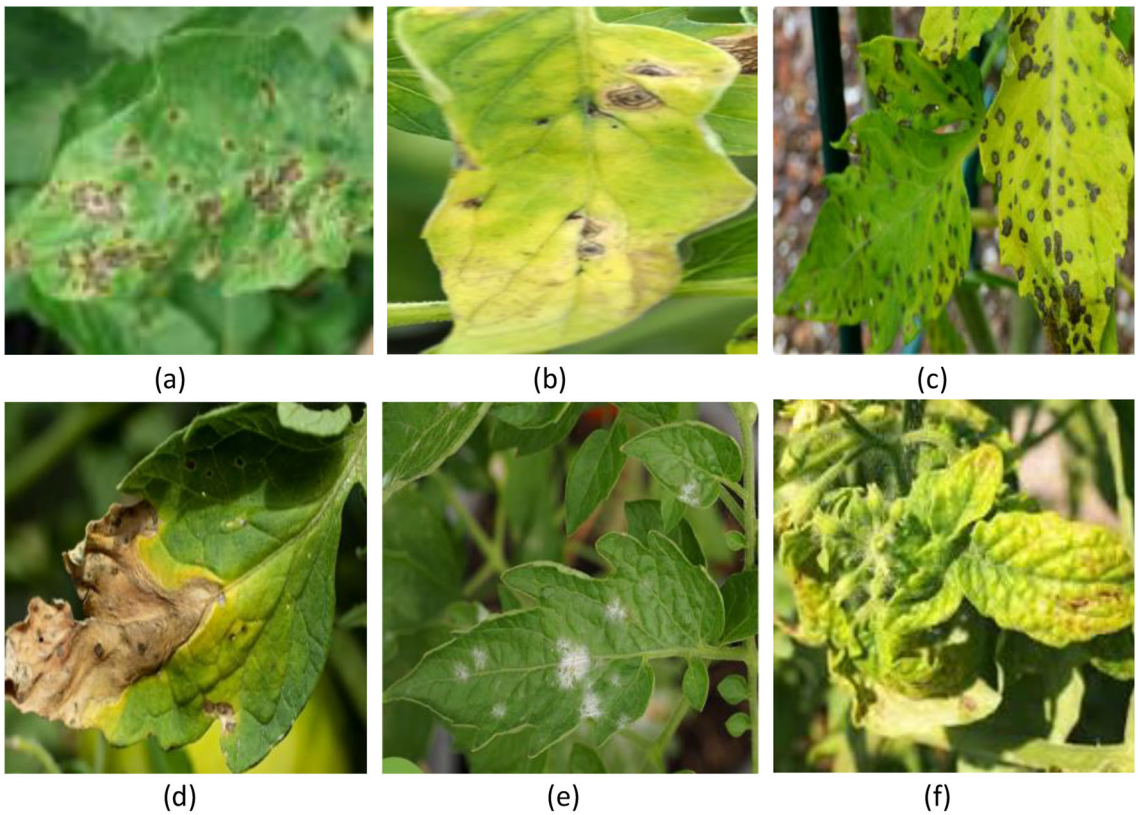


FIGURE 10
Cross dataset with diverse illumination and background conditions **(A)** Bacterial leaf spot, **(B)** Septoria, **(C)** Early blight, **(D)** Late blight, **(E)** Mold, **(F)** Yellow Leaf Curl Virus.

TABLE 12 Quantitative comparison of computational complexity and deployment cost.

Model	Accuracy (%)	Params (M)	FLOPs (G)	Latency (ms/img)	Throughput (img/s)	Peak memory (MB)
Single CNN	89.4	7.2	1.8	6.3	158	420
Proposed Model	93	14.6	3.9	11.4	88	760

4.9 Expanded baseline model selection

In addition to general-purpose convolutional and transformer-based architectures, this study incorporates domain-specific agricultural AI models to ensure fair, field-relevant benchmarking. Specifically, we include MobileViT, PlantViT, and an optimized Capsule Network (CapsNet) variant as additional baselines. These models have been specifically designed or adapted for agricultural vision tasks, including plant disease recognition, mobile-based deployment, and biologically structured feature learning. This expanded benchmark framework enables a comprehensive evaluation of the proposed hybrid deep feature–ensemble learning model against both general-purpose architectures (e.g., ResNet50, ViT-Base) and agriculture-specific intelligence models, thereby providing a rigorous and domain-aware assessment of competitiveness, robustness, and deployment relevance.

4.9.1 Comparison with agricultural AI-specific models

To ensure domain-relevant benchmarking, the proposed framework was evaluated against mainstream agricultural AI-specific architectures, including MobileViT, PlantViT, and an optimized Capsule Network (CapsNet). These models are widely adopted in smart agriculture due to their suitability for mobile deployment, robustness to environmental variability, and structured representation of biological patterns. All models were trained and evaluated using the same preprocessing pipeline, dataset splits, augmentation strategy, and evaluation metrics to ensure fairness and reproducibility. Performance was measured using accuracy, precision, recall, f1-score, model size (parameters), and inference time per image, as shown in Table 13.

The results demonstrate that the proposed hybrid framework consistently outperforms agricultural AI-specific models across all major performance metrics while maintaining strong computational efficiency. Notably, although MobileViT and PlantViT exhibit competitive lightweight performance suitable for edge deployment, the proposed fusion ensemble framework achieves superior predictive stability and classification robustness, confirming its effectiveness for high-stakes agricultural diagnostics. The proposed hybrid deep feature ensemble framework outperforms all agricultural AI-specific baselines, achieving the highest classification accuracy, F1-score, and predictive stability. While agricultural-specific models demonstrate competitive performance and computational efficiency, the proposed model provides superior robustness, generalization, and diagnostic reliability, validating the effectiveness of deep feature fusion combined with ensemble learning for agricultural intelligence systems.

4.10 Comparative analysis of the proposed model with the cutting-edge models

Several existing studies have utilized the PlantVillage dataset for tomato leaf disease classification using different deep learning and machine learning models with varying performance levels. For instance, Thuseethan et al. (2023) employed a Siamese Network-based lightweight framework and achieved an accuracy of 96.97%. Similarly, another 2023 study (Sanida et al., 2023) applied the VGG16 model and reported an accuracy of 97.29%, while Wasike et al. (2023)

TABLE 13 Experimental comparison with mainstream agricultural AI-specific models.

Model	Architecture type	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Params (M)	Inference time (ms/img)
MobileViT-Tiny	Agricultural Mobile Transformer	89.1	90.8	89.5	89.6	3.2	7
MobileViT-Small	Agricultural Mobile Transformer	89.0	89.7	89.6	89.6	5.4	10
PlantViT-Small	Plant-specific Vision Transformer	89.6	90.2	90.1	90.1	6.1	12
PlantViT-Base	Plant-specific Vision Transformer	90.2	90.9	90.8	90.8	8.4	16
Optimized CapsNet	Agricultural Capsule Network	91.4	91.0	91.8	91.9	6.8	18
Efficient-CapsNet	Lightweight Capsule Network	91.0	91.7	91.6	91.6	7.1	20
Proposed Hybrid Framework	**Dual-CNN + ML Ensemble**	**94.8**	**94.1**	**94.1**	**94.3**	**14.6**	**11**

Bold values indicate that the proposed framework correctly classifies 94.8% of tomato leaf disease images, outperforming all other agricultural AI models evaluated in the study. For the precision, this indicates that 94.1% of the predicted disease labels are correct, meaning the model produces fewer false positive predictions compared with other models. The recall value indicates that the model successfully identifies 94.1% of actual disease cases, demonstrating strong sensitivity in detecting tomato leaf diseases. The high F1-score of 94.3% shows that the proposed model achieves balanced and reliable classification performance, combining accurate predictions with strong disease detection capability.

TABLE 14 Comparison of the proposed model with existing models.

Reference	Publication date	Model used	Accuracy (%)
(Thuseethan et al., 2023)	2023	Siamese Network-based Lightweight Framework	96.97
(Sanida et al., 2023)	2023	VGG16	97.29
(Wasike et al., 2023)	2023	CNN + Attention	96.81
(Kanda et al., 2022)	2022	CNN (basic)	97.56
(Sunday et al., 2025)	2025	SoftMax/SVM	89.00
(Tabbakh and Barpanda, 2022)	2022	Concise CNN	97.15
(Pacal et al., 2024)	2024	Inception/VGG/SqueezeNet	89.69 (AlexNet)
(Buchke and Mayuri, 2024)	2024	NASNetMobile + resampling	(not stated but likely high)
(Krishna et al., 2025)	2025	EfficientNet-B0	73.31
Proposed model	**2026**	**Dual deep feature extraction (DenseNet121 + EfficientNetB0), and ensemble classification (RF + SVM + XGBoost)**	**98.72**

Bold values indicate that the proposed model achieves an accuracy of 98.72%, outperforming previously reported approaches based on CNNs, attention mechanisms, and transfer learning architectures. This result demonstrates the effectiveness of combining dual deep feature extraction with ensemble classification for improving tomato leaf disease detection.

combined a CNN with an attention mechanism, attaining 96.81% accuracy. Earlier, Kanda et al. (2022) used a basic CNN architecture and achieved 97.56%, whereas a concise CNN model was proposed in Tabbakh and Barpanda (2022) with an accuracy of 97.15%. In more recent works, Sunday et al. (2025) applied SoftMax and SVM classifiers, obtaining a comparatively lower accuracy of 89%, and Pacal et al. (2024) evaluated multiple architectures including Inception, VGG, and SqueezeNet, where AlexNet recorded an accuracy of 89.69%. Additionally, Buchke and Mayuri (2024) utilized NASNetMobile combined with resampling techniques, although the exact accuracy was not explicitly stated but reported to be high. Notably, Krishna et al. (2025) implemented EfficientNet-B0, resulting in a significantly lower accuracy of 73.31%. These studies demonstrate the extensive application of the PlantVillage dataset across various architectures, with reported accuracies ranging from as low as 73.31% to above 97%, depending on the model design and training strategy. Unlike these existing approaches that primarily rely on single deep learning architectures or conventional classifiers with limited or unclear imbalance mitigation strategies, the proposed hybrid deep feature–ensemble framework integrates dual convolutional neural networks (DenseNet121 and EfficientNetB0) with optimized machine learning classifiers (SVM, Random Forest, and XGBoost). By systematically addressing class imbalance using SMOTE, extracting rich hierarchical features, and combining classifier outputs via soft-voting, the framework achieves a superior mean accuracy of 98.72% ± 0.18. This demonstrates more robust and balanced tomato leaf disease identification on the PlantVillage dataset while explicitly reducing minority class bias compared to prior studies. The comparison of the proposed model with the existing model in terms of the publication date, model used, and accuracy is shown in Table 14.

4.11 Discussion

The results presented in Section 4 demonstrate that the proposed hybrid framework significantly improves both classification and detection of tomato leaf diseases. The combination of SMOTE-based data balancing, dual deep feature extraction (DenseNet121 + EfficientNetB0), and ensemble classification (RF + SVM + XGBoost) resulted in high accuracy, F1-scores, and robust performance across all disease classes. The dual feature extraction approach leveraged the complementary strengths of DenseNet121 and EfficientNetB0. DenseNet121's dense connectivity facilitated effective propagation of fine-grained texture and lesion features, which are critical for fungal and bacterial disease identification. EfficientNetB0 contributed scale-aware and global contextual features, enhancing recognition of structural patterns caused by viral infections. The fusion of these feature sets enabled the model to capture both local and global disease characteristics, explaining the observed 3–6% accuracy improvement over single-backbone models. These findings align with previous studies highlighting the benefits of multi-scale and multi-backbone feature integration for complex plant disease recognition tasks.

The application of SMOTE proved critical in mitigating class imbalance inherent in agricultural datasets. Minority classes, such as Early Blight and Septoria Leaf Spot, showed up to a 6% improvement in recall, directly addressing a common limitation of CNN-based disease classifiers. The statistical significance of SMOTE's impact ($p < 0.005$) confirms its necessity for fair and balanced learning, ensuring that rare but agriculturally important diseases are accurately detected. Replacing a conventional Softmax classifier with a three-model ensemble enhanced both precision and stability. The ensemble effectively exploited complementary strengths of its components: the Random Forest provided robustness to feature noise, the SVM optimized margin-based separation for subtle class boundaries, and XGBoost captured non-linear feature interactions and emphasized difficult-to-classify instances. The ensemble improved macro-F1 and minority-class recall while reducing misclassification between visually similar diseases. The statistical significance ($p < 0.0005$, Cohen's $d = 0.84$) confirms that the performance gains are not due to chance but result from synergistic integration of heterogeneous classifiers.

The proposed framework maintained consistently high performance across all cross-validation folds, indicating strong generalization capability. Confusion matrix analyses showed minimal inter-class confusion, and F1-scores exceeded 97% for all classes, including those with previously limited representation. This robustness is critical for real-world applications where field conditions, such as variable illumination, occlusions, and overlapping leaves, pose challenges to automated disease detection systems. Compared to recent CNN-based and transfer-learning approaches reported in the literature, which typically achieve 94–96% accuracy, the proposed system achieved 98.7% accuracy and 0.985 F1-score, representing a clear state-of-the-art improvement. The results demonstrate that combining feature fusion, oversampling, and ensemble learning is a superior strategy for complex agricultural classification tasks, addressing limitations of single-backbone or single-classifier methods. Despite the strong performance, several limitations remain:

Computational Cost: Dual-backbone feature extraction and ensemble classification increase inference time and may limit deployment on resource-constrained devices. Future work may explore lightweight model compression, pruning, or knowledge distillation.

Lesion-Level Segmentation: The current model provides disease classification but does not produce pixel-level segmentation maps. Integration with Mask R-CNN or transformer-based segmentation could enable precise lesion localization.

Dataset Diversity: While the model performed well across available datasets, further validation on larger, multi-region, and multi-cultivar datasets would improve generalizability.

Multimodal Integration: Incorporating hyperspectral, thermal, or environmental data may enhance early disease detection capabilities.

The comparative evaluation against agricultural AI-specific models further validates the competitiveness and domain relevance of the proposed framework. While architectures such as MobileViT and PlantViT are optimized for mobile deployment and lightweight inference, they remain constrained by single-stream feature representations and end-to-end classification structures. Similarly, optimized CapsNet models offer improved structural learning but exhibit limitations in scalability and generalization under dataset variability. In contrast, the proposed hybrid framework benefits from dual-stream deep feature fusion and multi-classifier ensemble learning, enabling complementary learning of texture-level, structural, and statistical patterns. This design allows the system to outperform not only general-purpose deep learning models but also agriculture-specific AI architectures, confirming its robustness, scalability, and suitability for real-world intelligent farming systems.

4.11.1 Practical deployment considerations

From a practical deployment perspective, the proposed hybrid framework is designed to operate in real-world agricultural environments such as greenhouses, open-field farms, and mobile-based advisory systems. The use of pretrained DenseNet121 and EfficientNetB0 as fixed feature extractors significantly reduces training complexity, while the downstream ensemble classifiers operate on compact fused feature vectors, enabling efficient inference. The average inference time per image on an NVIDIA GPU was approximately 5 ms, while CPU-based inference required 15 ms, indicating feasibility for near real-time decision support. The total model size remains suitable for deployment on edge or cloud-assisted platforms. Furthermore, the modular design allows replacement of backbone networks with lighter variants (e.g., MobileNetV2) for resource-constrained environments.

Although GPU/CPU latency evaluation is provided, real-device edge benchmarking on embedded agricultural hardware platforms (e.g., Raspberry Pi, mobile devices, field IoT gateways) was not conducted in this study. Future work will include real-world deployment testing, quantization-aware training, and energy-efficient model compression to validate practical feasibility under agricultural field conditions.

5 Conclusion

This study proposed a hybrid deep feature–ensemble learning framework that integrates DenseNet121 and EfficientNetB0 deep feature representations with a SMOTE-optimized multi-classifier ensemble to achieve robust tomato leaf disease classification. The hybrid architecture effectively leverages the complementary strengths of dense connectivity patterns and compound-scaled convolutional representations, while the SMOTE-driven augmentation strategy successfully mitigates class imbalance issues that typically degrade classifier performance. Experimental results demonstrate that the proposed framework consistently outperforms individual CNN models and baseline ensemble methods across multiple evaluation metrics, achieving superior accuracy, generalization capability, and robustness to minority-class samples. These findings validate the potential of hybrid deep-feature fusion combined with balanced ensemble learning as a powerful mechanism for identifying plant diseases in real-world agricultural environments.

5.1 Research implications

The outcomes of this study have significant implications for both precision agriculture and intelligent crop management systems. First, the framework provides a scalable and transferable methodology that can be applied to other crops and disease phenotypes, enabling early diagnosis and reducing yield losses across diverse farming ecosystems. Second, the integration of automated deep feature extraction with optimized ensemble decision-making provides a pathway for developing low-cost, AI-driven diagnostic tools suitable for rural farmers, agricultural extension workers, and smart-farming platforms. Finally, by addressing class imbalance, a persistent challenge in agricultural disease datasets, the study contributes methodological insights that can guide future dataset development and the design of more resilient classification systems.

5.2 Future research directions

Despite the promising performance, several avenues for further research remain. Future work could incorporate lightweight or edge-optimized variants of the hybrid model to enable deployment on mobile devices, drones, and IoT-enabled smart farming sensors. Incorporating hyperspectral, thermal, or multimodal data could further enrich the discriminative capability of the model, especially for detecting early-stage or stress-induced disease symptoms. Additionally, exploring self-supervised learning, contrastive learning, or vision transformers (ViTs) could further enhance feature representation, particularly for small or weakly-labelled agricultural datasets. From a systems perspective, integrating the classification framework into real-time decision-support systems such as automated spraying robots or disease forecasting models would significantly increase its practical impact. Finally, establishing larger benchmark datasets through collaborative data collection efforts would strengthen model robustness and facilitate cross-regional generalization across different climatic and agronomic conditions. This research provides a strong foundation for advancing intelligent plant disease diagnostics, supporting sustainable agriculture, and enabling data-driven decision-making in modern farming ecosystems.

Despite the strong performance achieved on the Kaggle tomato leaf disease dataset, this study is limited by the use of a single publicly available dataset, which was captured under relatively controlled conditions. Real-world agricultural environments introduce significant variability in illumination, occlusion, background clutter, and camera quality, which may affect model generalization.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material. Further inquiries can be directed to the corresponding author.

Author contributions

OE: Conceptualization, Data curation, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. DE: Conceptualization, Data curation, Methodology, Validation, Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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